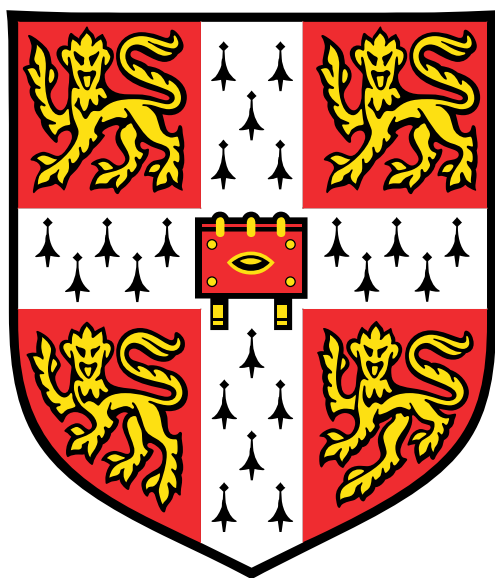


Hypercube Inequalities and Other Combinatorial Results



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Preface

This thesis is the result of my own work and includes nothing which is the outcome of work done in collaboration or content generated by artificial intelligence, except as declared in the preface and specified in the text. It is not substantially the same as any work that has already been submitted, or is being concurrently submitted, for any degree, diploma or other qualification at the University of Cambridge or any other University or similar institution except as declared in the preface and specified in the text. It does not exceed the prescribed word limit for the relevant Degree Committee.

Chapters 1, 2, 5, 6, and 7 are the fruits of independent work by the author.

Chapter 3 is joint work with Natalie Behague, Imre Leader, and Natasha Morrison.

Chapter 4 is joint work with Imre Leader.

Abstract

We solve problems in seven distinct areas of Combinatorics. First, we generalise the Harris-Kleitman inequality and answer a question of Kahn. Second, we consider posets whose largest antichain is a layer, continuing work by Harper to show that a Hamming ball has this property. Third, we prove there are induced subgraphs of a factorially large hypercube with any even girth which cannot embed into a hypercube edge layer. Fourth, in response to a paper by Fernández-Bretón, Sarmiento, and Vera, we prove that an abelian group with no element of order 4 can be countably coloured such that $\{2x, x + y, 2y\}$ is not monochromatic for any $x \neq y$ in the group. Fifth, we give an upper and lower bound for the maximal number of connected components we see in the plane upon intersecting polygonal regions with m and n sides and pin down the maximum if one polygon is convex. Sixth, we investigate Pólya's toroidal n -queens problem in multiple dimensions, proving there is an independent set of $n^{d-1} - O(n^{d-2})$ queens on $\mathbb{Z}_n^d$ if the least divisor of n exceeds $2^{\lfloor d/2 \rfloor + 1}$, but not n^2 queens if $d = 3$ and $n \equiv \pm 5, \pm 10 \pmod{25}$. Seventh, we build upon Boyd's Gossiping Problem, determining the most number of gossip items that can be known in total when the number of calls is restricted.

Introduction

In mathematics, the subfield of Combinatorics answers elementary questions with uncontrived, spontaneous proof. This notion appears to differ from deeply theoretical research in much of pure mathematics, even from that which starts out with a combinatorial problem. However, for our purposes, Combinatorics is defined by its vast scope, interrelated with discussions of groups, geometry, mathematical games, statistics, and computation. The joy of Combinatorics consists in artless curiosity and uncomplicated reason, cultivated without clinging to theories that necessitate a specialised approach to problem-solving. There is virtue in collecting a variety of enquiries that showcase versatile intelligence with interdisciplinary scope.

Disjointed as the work of resolving a breadth of conjectures and enquiries may seem, the topics visited in this thesis are to an extent connected by the key word *hypercube*. Chapter 1 describes an upward closed event in terms of a family of sets in Q_n , the power set of a set with n elements. Chapter 2 concerns the Hamming ball, a subfamily of Q_n that contains the sets arising from an m -element set by removing or including at most r elements. Chapter 3 investigates a graph with node set Q_n whose edges connect sets that differ by a singleton, commonly known as the hypercube graph Q_n . The further chapters do not so much as mention the hypercube, for Chapter 4 studies infinite groups, Chapter 5 investigates polygonal regions, Chapter 6 examines toroidal grids, and Chapter 7 happens to elaborate on a local undergraduate challenge problem about Gossiping.

The first chapter explains how the Harris-Kleitman inequality could be generalised. It concerns upward closed families X of subsets of a finite set with n elements, those for which whenever $S \in X$ and $S \subset T$, also $T \in X$. For upward closed systems X and Y , the Harris-Kleitman inequality states that the correlation between a uniformly random set being in X and its being in Y is non-negative: if $|X| = \alpha 2^n$ and $|Y| = \beta 2^n$, then $|X \cap Y| \geq \alpha \beta 2^n$. We give a proof of this inequality that sheds new light on a conjecture of Richards and Sahi concerning three upward closed systems. Their conjecture would imply that for upward closed systems X , Y , and Z , each of size $\alpha 2^n$, if $\alpha \leq \frac{1}{3}$, then at most $\frac{4}{9} \cdot 2^n$ sets occur in exactly one of them. Although Kahn conjectured this to be true for any α , we show this not to be the case in the range $\frac{1}{3} < \alpha < \frac{3}{7}$. Furthermore, in response to Gladkov's resolution of a question due to Steif, we state and prove a new linear correlation inequality for k upward closed systems, where $k \geq 4$.

The second chapter describes $\text{width}(P)$, the size of the largest antichain contained in a finite poset P , where P is a Hamming ball. Partitioning a finite poset P into layers P_k according to the length of the longest chain terminating at an element x yields a lower bound $\text{width}(P) \geq |P_k|$

for all k . By Sperner's theorem, the width of $P = (Q_n, \subseteq)$ is in fact its largest layer size. In general, if there is a collection $\mathcal{C}$ of c chains containing each $x \in P_k$ exactly $\frac{c}{|P_k|}$ times, known as a unit flow, it follows that across elements of an antichain, $\sum_x |P_k|^{-1} \leq 1$. This inequality, for $P = (Q_n, \subseteq)$, is independently due to Yamamoto, Meshalkin, Bollobás, and Lubell, who deduced Sperner's theorem from it. Kleitman showed that conversely, if this inequality holds, then a unit flow exists, where $\mathcal{C}$ could be a multiset. Building on these ideas, Harper discovered that the class of posets $P = \bigcup_{k=0}^{\infty} P_k$ with a unit flow that satisfy $|P_{k-1}| \cdot |P_{k+1}| \leq |P_k|^2$ for all k are closed under taking products. We continue this enquiry to prove there is a unit flow in Hamming balls.

The third chapter discusses the possible subgraphs of the hypercube graph Q_n and its n edge layers. Havel and Moravek studied the condition for when a graph can be realised as a subgraph of Q_n . Axenovich, Martin, and Winter noticed that there is a similar condition for when a graph can be realised as a subgraph of an edge layer in Q_n . They proved that a graph of girth 8 exists that is a subgraph of Q_n , but not of any edge layer. We prove their conjecture that a graph of arbitrarily large girth can have this property, even if we require it to be an induced subgraph of Q_n .

The fourth chapter elaborates on a question due to Fernández-Bretón, Sarmiento and Vera. They showed that whenever a direct sum of $\boxplus_{\omega}(\aleph_0)$ or more copies of $\mathbb{Z}_4$, the cyclic group of order 4, is countably coloured, there are arbitrarily large finite sets X whose sumsets

$$X + X = \{x_1 + x_2 : (x_1, x_2) \in X \times X\}$$

are monochromatic. They asked if the elements of order 4 are necessary, in the following strong sense: if G is an abelian group with no element of order 4, such as direct sums of the Prüfer group $\mathbb{Z}_{p^\infty} = \bigcup_{k=1}^{\infty} \mathbb{Z}_{p^k}$ (p is an odd prime, $\mathbb{Z}_n$ is the multiplicative group of n -th complex unit roots), can there always be a countable colouring of G for which there is not even a set $\{x, y\}$ of size 2 with a monochromatic sumset? We answer this question in the affirmative.

The fifth chapter addresses a question of Croft about intersecting two polygonal regions in the plane. The intersection of a triangle and a triangle is convex, hence connected, but intersecting two quadrilaterals, there could be as many as four connected components, permitting concave polygons. We bound the maximal possible number $f(m, n)$ of connected regions in the intersection of an m -gonal region and an n -gonal region, showing that $f(5, 5) = 5$ and

$$\left\lfloor \frac{m}{2} \right\rfloor \cdot \left\lfloor \frac{n}{2} \right\rfloor \leq f(n, m) \leq \left\lfloor \frac{m}{2} \right\rfloor \cdot \frac{n}{2} + \frac{m}{2}.$$

Furthermore, if the m -gon is constrained to be convex, then we can prove that the maximal number of connected regions is $\max\left(\left\lfloor \frac{m+n-2}{2} \right\rfloor, n-2\right)$.

The sixth chapter studies the independence number of multidimensional chess graphs. In such a graph, the vertices are the fields of a board, such as the $n \times n$ chessboard or the toroidal board $\mathbb{Z}_n \times \mathbb{Z}_n$. We focus on the queens graph, where one field is joined by an edge to another if a queen can move from one field to the other in chess. A well-known problem from the 19th century asks to find an independent set of size n for queens on the $n \times n$ chessboard. Pólya showed that the independence number of the queens graph on $\mathbb{Z}_n^2$ can be n only if n is not a multiple of 2 or 3. Monsky found the independence number to be n , $n - 1$, or $n - 2$ if $n \equiv \pm 1 \pmod{6}$, $n \equiv \pm 2 \pmod{12}$, or otherwise, respectively. We replicate his construction and consider its ramifications in d dimensions. If n has no prime divisor less than 2^d , we demonstrate an independent set of n^{d-1} queens on $\mathbb{Z}_n^d$, as well as an independent set of $n^{d-1} - O(n^{d-2})$ queens if n has no prime divisor less than $2^{\lfloor d/2 \rfloor + 1}$, which we conjecture to be possible for all n . Furthermore, we show that n^2 independent queens on $\mathbb{Z}_n^3$ are impossible for $n \equiv \pm 5, \pm 10 \pmod{25}$.

The seventh chapter solves a problem related to Boyd's Gossiping Problem. There are n people who each know an item of gossip. In a telephone call, two people share all the gossip they know. How many calls are needed for all of them to be informed of all the gossip? One strategy would be for don 1 to call the dons $2, 3, \dots, n$ in this order twice. If each don counts how many items of gossip she knows, this is a strategy that maximises the total count with every call. In the Gossiping Problem, our greedy strategy is not optimal, for the total score is n^2 only after $2n - 3$ calls, whereas if $n \geq 4$, there is a strategy with $2n - 4$ calls. We initiate and solve the related Greedy Gossiping Problem: given a fixed number $m < 2n - 4$ of calls, what is the maximal value of the total gossip count? We show that surprisingly, the greedy strategy is optimal.

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Chapter 1

Multivariate Correlation Inequalities

1.1 Introduction

Let $(P, <)$ be a poset, which we refer to as P . If $X \subseteq P$, let $\mu(X) = \frac{|X|}{|P|}$ be the *density* of X .

We say that $X \subseteq P$ is an *upward closed* system if whenever $x < y$ and $x \in X$, also $y \in X$. A *downward closed* system is an upward closed system in the reverse poset $(P, >)$. For instance, the complement of X , $X^c = P \setminus X$, is downward closed.

Let $(Q_n, \subseteq)$ be the power set of a set with n elements, ordered by the subset relation. Specifically, we use the definition $0 = \emptyset$, $n + 1 = n \cup \{n\}$ from set theory, so Q_n is the power set of n .

Let X and Y be upward closed systems in Q_n , and let $S \in Q_n$ be sampled uniformly at random. Then the correlation between the events $S \in X$ and $S \in Y$ is non-negative, by the following result.

Theorem 1.1.1 (Harris [1], Kleitman [2]). *If X and Y are upward closed systems in Q_n , then*

$$\mu(X \cap Y) - \mu(X)\mu(Y) \geq 0.$$

If $S \in X$ and $S \in Y$ are independent, we say that X and Y are independent. For $t \in n$, we say that X depends on t if the events $t \in S$ and $S \in X$ are not independent. By evaluating the case of equality in Theorem 1.1.1, we deduce the following.

Corollary 1.1.2. *The upward closed systems X and Y in Q_n are independent if and only if there is no $t \in n$ such that X depends on t and Y depends on t .*

Another way to view Theorem 1.1.1 is that given $\mu(X)$ and $\mu(Y)$, it determines the minimal size of $\mu(X \cap Y)$, and hence the maximal size of $\mu(X \cap Y^c)$ and $\mu(X^c \cap Y)$.

More generally, for $k \geq 3$, let $X_1, X_2, \dots, X_k$ are upward closed systems in Q_n of given densities. How large can the 2^k possible intersections of each system or its complement be? If these systems are pairwise independent, the density of such an intersection is given by a product. For example,

$$\mu(X_1 \cap X_2^c \cap X_3^c) = \mu(X_1) \cdot (1 - \mu(X_2)) \cdot (1 - \mu(X_3))$$

for three independent systems. Are there any *correlation inequalities*, i.e. algebraic inequalities like Theorem 1.1.1 [5], in which the case of equality is when the k systems are independent?

Let X, Y, Z be three upward closed systems in Q_n with equal density α . For $0 \leq i \leq 3$, define U_i to contain the sets in Q_n that are in exactly i of the systems. Now if the systems are independent, $\mu(X \cap Y^c \cap Z^c) = \alpha(1 - \alpha)^2$, which is maximised by $\alpha = \frac{1}{3}$. Kahn asked the following questions.

Question 1.1.3 (Kahn [3]). *Does the maximum of $\mu(U_i)$ equal $\frac{4}{9}$, the maximum of $3\alpha(1 - \alpha)^2$? If not, can $X \cap Y^c \cap Z^c$, $X^c \cap Y \cap Z^c$, $X^c \cap Y^c \cap Z$ each be of density more than $\frac{4}{27}$?*

Surprisingly, our main result shows that the answer to the first part of Question 1.1.3 is no.

Theorem 1.1.4. *Let $\alpha = \frac{a}{2^m}$ be strictly between $\frac{1}{3}$ and $\frac{3}{7}$. Then for sufficiently large n , there are three upward closed systems $X, Y, Z \subseteq Q_n$ of density α where the density of U_1 exceeds $\frac{4}{9}$.*

Depending on α , for $\alpha < \frac{1}{3}$, if the following holds, then $3\alpha(1 - \alpha)^2$ is the maximum of $\mu(U_1)$.

Conjecture 1.1.5 (Richards [4], Sahi [5]). *For upward closed systems $X, Y, Z \subseteq Q_n$,*

$$2\mu(X \cap Y \cap Z) + \mu(X)\mu(Y)\mu(Z) \geq \mu(X)\mu(Y \cap Z) + \mu(Y)\mu(Z \cap X) + \mu(Z)\mu(X \cap Y).$$

Originally stated as a theorem [4] (but proven incorrectly), Conjecture 1.1.5 is the special case of a correlation inequality for k upward closed systems conjectured by Sahi [5].

Theorem 1.1.6. *If Conjecture 1.1.5 holds, then for upward closed systems $X, Y, Z \subseteq Q_n$ of equal density α , $\mu(U_1) \leq \frac{3\alpha(1-\alpha^2)}{3\alpha+1}$ if $\alpha > \frac{1}{3}$ and $\mu(U_1) \leq 3\alpha(1 - \alpha)^2$ if $\alpha \leq \frac{1}{3}$.*

Since the maximum of $\frac{3\alpha(1-\alpha^2)}{3\alpha+1}$ is approximately 0.45921, if we could find X, Y, Z for which $X \cap Y^c \cap Z^c$, $X^c \cap Y \cap Z^c$, $X^c \cap Y^c \cap Z$ have density more than 0.15308, we would thereby disprove Conjecture 1.1.5 [3]. However, it could be that our examples which prove Theorem 1.1.4 are in fact optimal for any fixed α , reducing the maximum of $\mu(U_1)$ to approximately 0.44748.

The first proven correlation inequality that is not a simple consequence of Theorem 1.1.1 was found by Gladkov [6] to answer a question of Steif [7].

Theorem 1.1.7 (Gladkov [6]). *Let $X_1, X_2, \dots, X_k, V, W$ be a partition of Q_n where $X_i \cup W$ is an upward closed system for $i = 1, 2, \dots, k$. Then*

$$\mu(V)\mu(W) \geq \sum_{1 \leq i < j \leq k} \mu(X_i)\mu(X_j).$$

If we let X_1, X_2, X_3 in Theorem 1.1.7 be $X \cap Y^c \cap Z^c, X^c \cap Y \cap Z^c, X^c \cap Y^c \cap Z$ from Question 1.1.3, we obtain a further constraint on their possible densities. We conjecture an adaptation of this constraint, one that would show the answer to the second part of Question 1.1.3 to be no.

Conjecture 1.1.8. *Let $X_\emptyset, X_1, X_2, X_3, X_{12}, X_{13}, X_{23}, X_{123}$ be a partition of Q_n where the four parts including i as an index form an upward closed system of density α ($i = 1, 2, 3$). Then*

$$\mu(X_1)\mu(X_2) + \mu(X_1)\mu(X_3) + \mu(X_2)\mu(X_3) \leq \frac{16}{243}.$$

The proof of Theorem 1.1.7 proceeds by induction on n [6], similarly to the standard proof of Theorem 1.1.1 [2]. In search for a proof of Conjecture 1.1.5, we present a different new proof of Theorem 1.1.1 that inducts on how many elements both systems depend on. Furthermore, in response to the degree k polynomial inequalities conjectured by Sahi [5], we show that an extremal value for the partial sums of the Inclusion-Exclusion formula is achieved by independent systems.

Theorem 1.1.9. *Let $X_1, \dots, X_k \subseteq Q_n$ be upward closed systems and let $\mu(X_i) = \alpha_i$ be fixed for $i = 1, \dots, k$. For $2 \leq l \leq k$, the following sum is minimised by pairwise independent $X_1, \dots, X_k$:*

$$M_{l,k} = \sum_{j=l}^k (-1)^{j-l} \sum_{I \subseteq \{1,2,\dots,k\}, |I|=j} \mu\left(\bigcap_{i \in I} X_i\right).$$

For sake of completeness, we begin by reproving Theorem 1.1.1 and Corollary 1.1.2. After that, we turn to constructing the family of examples that prove Theorem 1.1.4, and we demonstrate a conditional upper bound, Theorem 1.1.6. Finally, we prove Theorem 1.1.9.

1.2 A novel proof of Theorem 1.1.1 and Corollary 1.1.2

It is well-known that Theorem 1.1.1 and the more general finite FKG inequality [8] can be proven by induction on n [9] or by coupling Markov chains [10]. We provide a seemingly unprecedented purely combinatorial proof of Theorem 1.1.1 and analyse its relevance to Conjecture 1.1.5.

Proof of Theorem 1.1.1 and Corollary 1.1.2. We reduce the inequality to its case of equality with so-called *exchange operators*, by making the $t \in n$ that both of $X, Y \subseteq Q_n$ depend on disappear.

By definition, X does not depend on t when $\{S \in X | t \in S\}$ contains exactly half the sets in X . Since X is upward closed, $S \in X$ whenever $S \setminus \{t\} \in X$, and so $\{S \in X | t \in S\}$ contains at least half the sets in X . Hence, X does not depend on t if and only if $S \in X \Leftrightarrow S \setminus \{t\} \in X$ for all S .

Let D_X be the set of $t \in n$ that X depends on, and similarly for Y . Since removing an element $t \notin D_X$ does not change whether $S \in X$, it follows that $S \in X$ if and only if $S \cap D_X \in X$. Now if there is no t for which X depends on t and Y depends on t , then $D_X \cap D_Y = \emptyset$. Hence, the pairs of sets (S, T) for which $S \cap D_X \in X$ and $T \cap D_Y \in Y$ are in bijection with the pairs of sets (S', T') for which $S' \cap D_X \in X$ and $S' \cap D_Y \in Y$, via switching $S \cap D_Y$ and $T \cap D_Y$. Since these conditions are equivalent with S being in X, Y , both, or neither, we see $|X| \cdot |Y| = |X \cap Y| \cdot 2^n$.

We define exchange operators E_{tu} , a symmetric version of the compression operator C_{ut} used in proving the Kruskal-Katona theorem [11]. Let $Z \subseteq Q_n$ and $t, u \in n$. Then

$$E_{tu}(Z) = \{S \triangle \{t, u\} | S \in Z\}.$$

Suppose that X does not depend on u and Y does not depend on t . Let us classify the sets $S \in Q_n$ into four levels, according to whether $t \in S$ or $u \in S$. We label $S \in Q_n$ with $S \cup \{t, u\}$, identifying the sets up to elements t and u . Let X_0 contain the labels of the $S \in X$ for which $S \setminus \{t\} \notin X$ and X_1 contain the rest of the labels of the $S \in X$. Similarly, define Y_0 and Y_1 with respect to Y and u .

Lemma 1.2.1. $|X \cap E_{tu}(Y)| - |X \cap Y| = |X_0 \cap Y_0| \geq 0$.

Proof. In the level of $S \in Q_n$ that contain both or neither t and u , the same sets are in Y and $E_{tu}(Y)$, so $|X \cap E_{tu}(Y)| - |X \cap Y|$ only depends on the levels considered in the following table.

Level	Label set of X	Label set of Y	Label set of $E_{tu}(Y)$
$t \in S, u \notin S$	$X_0 \sqcup X_1$	Y_1	$Y_0 \sqcup Y_1$
$t \notin S, u \in S$	X_1	$Y_0 \sqcup Y_1$	Y_1

In each level, the possible labels of a set S in X or in Y are determined, for X does not depend on u and Y does not depend on t . By swapping t and u , we obtain the labels for $E_{tu}(Y)$. Thus, $|X \cap E_{tu}(Y)| - |X \cap Y|$ is the difference between $|(X_0 \sqcup X_1) \cap (Y_0 \sqcup Y_1)| + |X_1 \cap Y_1|$ and $|(X_0 \sqcup X_1) \cap Y_1| + |X_1 \cap (Y_0 \sqcup Y_1)|$, which is $|X_0 \cap Y_0|$. $\square$

We prove by induction on $d = |D_X \cap D_Y| > 0$ that $|X \cap Y| > |X| \cdot |Y| \cdot 2^{-n}$ – as we have seen, equality holds when $d = 0$. Let X and Y be upward closed systems in Q_n with d elements both depend on. If both X and Y depend on t , let u be an element neither depends on – if no such element u exists among the n elements, we can append another element without changing the density of the systems. Then $E_{tu}(Y)$ depends on u , not on t , and so there are $d - 1$ elements both X and $E_{tu}(Y)$ depend on.

By Lemma 1.2.1 for X and $E_{tu}(Y)$, $|X \cap Y| \geq |X \cap E_{tu}(Y)|$. Also, if $d = 1$, we will show this inequality to be strict. Consider X and Y with $D_X \cap D_Y = \emptyset$. Then $S \in X_0$ if and only if $S \cap D_X \in X_0$, and similarly for Y . Hence, like before, $|X_0 \cap Y_0| = |X_0| \cdot |Y_0| \cdot 2^{-n}$. If X depends on t and Y depends on u , then X_0 and Y_0 are not empty, so $|X_0 \cap Y_0| > 0$, as required. This proves Corollary 1.1.2, as well as Theorem 1.1.1. $\square$

Let us explore how this method could prove Conjecture 1.1.5, i.e.

$$2\mu(X \cap Y \cap Z) - \mu(X)\mu(Y \cap Z) - \mu(Y)\mu(X \cap Z) - \mu(Z)\mu(X \cap Y) \geq -\mu(X)\mu(Y)\mu(Z)$$

for any upward closed systems $X, Y, Z \subseteq Q_n$. If Z does not depend on an element X or Y depends on, then the densities $\mu(X \cap Y \cap Z) = \mu(X \cap Y)\mu(Z)$, $\mu(X \cap Z) = \mu(X)\mu(Z)$, and $\mu(Y \cap Z) = \mu(Y)\mu(Z)$, so the result follows from Theorem 1.1.1. Thus, it would suffice to show that whenever Z does not depend on t , whilst X and Y do not depend on u , replacing Z with $E_{tu}(Z)$ cannot decrease the left-hand side.

Like before, among sets S with $t, u \in S$, define X_0 and X_1 , Y_0 and Y_1 , $(X \cap Y)_0$ and $(X \cap Y)_1$ according to whether $S \setminus \{t\}$ is in the system, as well as Z_0 and Z_1 , with respect to u . Now

$$\begin{aligned} (X \cap Y)_0 &= (X_0 \sqcup X_1) \cap (Y_0 \sqcup Y_1) \setminus (X \cap Y)_1 \\ &= (X_0 \cap Y_0) \sqcup (X_1 \cap Y_0) \sqcup (X_0 \cap Y_1). \end{aligned}$$

By Lemma 1.2.1, replacing Z with $E_{tu}(Z)$, the density $\mu(X \cap Y \cap Z)$ increases by $\mu((X \cap Y)_0 \cap Z_0)$, $\mu(X \cap Z)$ by $\mu(X_0 \cap Z_0)$, and $\mu(Y \cap Z)$ by $\mu(Y_0 \cap Z_0)$. Hence, the left-hand side changes by

$$2\mu((X_0 \cap Y_0 \cap Z_0) \sqcup (X_1 \cap Y_0 \cap Z_0) \sqcup (X_0 \cap Y_1 \cap Z_0)) - \mu(X)\mu(Y_0 \cap Z_0) - \mu(Y)\mu(X_0 \cap Z_0).$$

While X and X_1 are upward closed sets, in general, their difference X_0 is not upward closed. Hence, the change could be negative, unless Z is independent from X and Y . To prove or disprove Conjecture 1.1.5, one could study the sum of these changes in a suitable order.

1.3 Weighted posets and Theorem 1.1.4

To answer Question 1.1.3, we develop some theory to begin with.

Lemma 1.3.1. *If $X, Y \subseteq P$ are upward closed systems in $(P, <)$, then so are $X \cap Y$ and $X \cup Y$.*

Proof. If $x \in P$ is an element of $X \cap Y$, then $x \in X$ and $x \in Y$. Hence, if $x < y$ in P , then $y \in X$ and $y \in Y$, so $y \in X \cap Y$.

If $x \in P$ is an element of $X \cup Y$, then either $x \in X$ or $x \in Y$. If $x < y$ in P , then $y \in X$ or $y \in Y$ follows, and so $y \in X \cup Y$. $\square$

Lemma 1.3.2. *Let $X \subset X'$ be distinct upward closed systems in $(P, <)$. In this case, there is an upward closed system X'' with $X \subset X'' \subseteq X'$ and $|X''| = |X| + 1$.*

Proof. If x is a maximal point in $X' \setminus X$, then $X'' = X \cup \{x\}$ is upward closed. Indeed, this way, if $x < y$, then $y \notin X' \setminus X$, but $y \in X'$, because X' is upward closed. Hence, $y \in X$. $\square$

Corollary 1.3.3. *For all integers a and $m \geq 1$ with $\alpha = \frac{a}{2^m} \in [0, 1]$, there is an upward closed system $X \subseteq Q_{n_\alpha}$ with density α .*

Proof. We induct on a with $X = \emptyset$ for $a = 0$ and X'' for $a+1$, by Lemma 1.3.2 with $X' = Q_m$. $\square$

These preliminaries enable us to prove our main result.

Proof of Theorem 1.1.4. By Corollary 1.3.3, for any binary fractions α, β , and γ , there are upward closed systems $X \subseteq Q_{n_\alpha}$, $Y \subseteq Q_{n_\beta}$, and $Z \subseteq Q_{n_\gamma}$ with density α, β , and γ , respectively. For $n = n_\alpha + n_\beta + n_\gamma$, these can be viewed as independent upward closed systems in Q_n . Thus, $\mu(X \cap Y \cap Z) = \alpha\beta\gamma$, and similarly for complements of X, Y, Z , whence

$$\mu(U_1) = \alpha(1 - \beta)(1 - \gamma) + (1 - \alpha)\beta(1 - \gamma) + (1 - \alpha)(1 - \beta)\gamma.$$

By Lemma 1.3.1, $X' = X \cup (Y \cap Z) = X \sqcup (X^c \cap Y \cap Z)$ is an upward closed system. Let us replace X with X'' , where $X \subseteq X'' \subseteq X'$. U_1 does not change, because $Y \cap Z$ and U_1 are disjoint. Hence, if X' is larger than Y and Z , then by Lemma 1.3.2, the size of X can be incremented until it reaches the size of the largest system. Proceeding in the same way with Y and Z , we obtain three systems whose densities are equal, without ever changing $\mu(U_1)$.

Suppose that $\alpha \geq \beta \geq \gamma$. As we have explained, our choice of independent $X, Y, Z \subseteq Q_n$ generates an example with equal density α if and only if $\gamma + (1 - \gamma)\alpha\beta \geq \alpha$ and $\beta + (1 - \beta)\alpha\gamma \geq \alpha$. If we multiply $\beta \geq \gamma$ by $(1 - \alpha)$, that rearranges to $\gamma + (1 - \gamma)\alpha\beta \leq \beta + (1 - \beta)\alpha\gamma$, so it suffices to guarantee the first condition, equivalent with $\gamma \geq \frac{\alpha(1-\beta)}{1-\alpha\beta}$ and $\alpha \leq \frac{\gamma}{1-\beta+\beta\gamma}$.

Suppose that $\alpha = \beta$. Then with $\gamma = \frac{\alpha(1-\alpha)}{1-\alpha^2} = \frac{\alpha}{1+\alpha}$, the density of U_1 would be

$$\mu(U_1) = \gamma(1 - \alpha)^2 + 2(1 - \gamma)\alpha(1 - \alpha) = \gamma(1 - \alpha)(1 - 3\alpha) + 2\alpha(1 - \alpha),$$

which evaluates to $\frac{\alpha(1-\alpha)(3-\alpha)}{1+\alpha}$. The maximum of this expression over $\alpha \in [0, 1]$, by computer algebra, is at $\alpha \approx 0.370556$, with $\mu(U_1) \approx 0.447485$. This exceeds $\frac{4}{9} \approx 0.4444$.

Regarding possible densities α with which $\mu(U_1) > \frac{4}{9}$, in this example, $\alpha \in (\frac{1}{3}, 0.409]$. We can extend this range up to $\frac{3}{7}$ with examples where $\beta = \gamma$. In this case, for $\alpha = \frac{\gamma}{1-\gamma+\gamma^2}$, $\mu(U_1) > \frac{4}{9}$ holds if $0.284 < \gamma < \frac{1}{3}$, where α ranges from 0.357 to $\frac{3}{7}$.

Since binary fractions are dense in $[0, 1]$, in practice, we can choose γ to approximate $\frac{\alpha}{1+\alpha}$ or α to approximate $\frac{\gamma}{1-\gamma+\gamma^2}$ with arbitrary precision. By continuity, it follows that for any binary fraction α in the range $\frac{1}{3} < \alpha < \frac{3}{7}$, it is possible to accomplish $\mu(U_1) > \frac{4}{9}$. $\square$

Corollary 1.3.4. *The answer to Question 1.1.3 is no whenever $n \geq 11$.*

Proof. In our counterexample, we specify $\alpha = \frac{3}{8}$, $\beta = \frac{5}{16}$, $\gamma = \frac{5}{16}$, generating independent $X, Y, Z \subseteq Q_{3+4+4} = Q_{11}$ of these densities. Now $\mu(X \cap Y^c \cap Z) = \frac{3}{8} \cdot \frac{11}{16} \cdot \frac{5}{16} > \frac{1}{16}$, so the systems Y and Z can be increased up to the size of X , while the density of U_1 remains $\frac{913}{2048} > 0.4458$. $\square$

Note that our construction is not a counterexample to Conjecture 1.1.5. If X, Y, Z are independent,

$$2\mu(X \cap Y \cap Z) - \mu(X)\mu(Y \cap Z) - \mu(Y)\mu(Z \cap X) - \mu(Z)\mu(X \cap Y) + \mu(X)\mu(Y)\mu(Z) = 0,$$

so equality holds. Our key idea was to move elements from $X^c \cap Y \cap Z$ into $X \cap Y \cap Z$, and similarly for Y and Z . Clearly, this does not change Y or Z , but it increases intersections of X by a density δ . Hence, our move increases the difference of the two sides in Conjecture 1.1.5 by

$$(2 - \mu(Y \cap Z) - \mu(Y) - \mu(Z) + \mu(Y)\mu(Z))\delta,$$

which is the sum of $(1 - \mu(Y \cap Z))\delta \geq 0$ and $(1 - \mu(Y))(1 - \mu(Z))\delta \geq 0$. Therefore, when we form systems this way, we can only obtain systems for which Conjecture 1.1.5 is true.

1.4 An upper bound for $\mu(U_1)$

Suppose that $X, Y, Z \subseteq Q_n$ are of equal density α . Let U_i be the set of $S \in Q_n$ contained in exactly i of the systems X, Y, Z , and $\mu(U_i) = u_i$. Then Conjecture 1.1.5 simplifies to

$$2 \cdot \mu(U_3) + \alpha^3 \geq \alpha \cdot (\mu(U_2) + 3\mu(U_3))$$

because $\mu(Y \cap Z) = \mu(X^c \cap Y \cap Z) + \mu(X \cap Y \cap Z)$ etc.

Corollary 1.4.1. *If Conjecture 1.1.5 is true, then whenever $X, Y, Z \subseteq Q_n$ have density α ,*

$$(3\alpha - 1)u_0 + (1 - \alpha)^3 \geq (1 - \alpha)u_1.$$

Proof. Since $X^c, Y^c, Z^c \subseteq Q_n$ are upward closed systems of density $1 - \alpha$ in the reverse poset of Q_n , which is identical to Q_n , we obtain this bound involving $\mu(U_0)$ and $\mu(U_1)$. $\square$

Let us now deduce Theorem 1.1.6, which suggests the answer to Question 1.1.3 would be yes if the density of each system were at most $\frac{1}{3}$.

Proof of Theorem 1.1.6. By double-counting $\mu(Q_n)$ and $\mu(X) + \mu(Y) + \mu(Z)$,

$$1 = u_0 + u_1 + u_2 + u_3, \quad 3\alpha = u_1 + 2u_2 + 3u_3.$$

Suppose that $\alpha > \frac{1}{3}$, so the coefficient $3\alpha - 1 > 0$. Substituting the estimate for u_0 from Corollary 1.4.1, as well as the trivial $u_2 \geq 0$, into the equation $3(1 - \alpha) = u_2 + 2u_1 + 3u_0$, we find

$$\begin{aligned} 3(1 - \alpha) &\geq 0 + 2u_1 + 3 \cdot \frac{(1 - \alpha)u_1 - (1 - \alpha)^3}{3\alpha - 1}, \\ 3(1 - \alpha)(3\alpha - 1) &\geq (6\alpha - 2)u_1 + (3 - 3\alpha)u_1 - 3(1 - \alpha)^3, \\ 3(1 - \alpha)[3\alpha - 1 + (1 - \alpha)^2] &\geq (3\alpha + 1)u_1, \\ \frac{3\alpha(1 - \alpha^2)}{3\alpha + 1} &\geq u_1. \end{aligned}$$

Suppose that $\alpha \leq \frac{1}{3}$, so that $3\alpha - 1 < 0$. Then $(1 - 3\alpha)u_0 + (1 - \alpha)u_1 \leq (1 - \alpha)^3$, which, knowing that $u_0 \geq (1 - \alpha)^3$, additionally informs us that $(1 - \alpha)u_1 \leq 3\alpha u_0$. Therefore,

$$\begin{aligned} (1 - 3\alpha) \frac{1 - \alpha}{3\alpha} u_1 + (1 - \alpha)u_1 &\leq (1 - \alpha)^3, \\ (1 - 3\alpha)u_1 + 3\alpha u_1 &\leq 3\alpha(1 - \alpha)^2, \\ u_1 &\leq 3\alpha(1 - \alpha)^2, \end{aligned}$$

which is the claimed upper bound. $\square$

1.5 A linear correlation inequality for k systems

Before proving Theorem 1.1.9, let us rewrite $M_{l,k}$ in terms of the symmetric sets U_i .

Lemma 1.5.1. *Let $X_1, \dots, X_k \subseteq Q_n$ be upward closed systems and U_i be the set of elements $S \in Q_n$ contained by exactly i of the systems. Then*

$$M_{l,k} = \sum_{i=l}^k \binom{i-1}{l-1} \mu(U_i).$$

Proof. Recall that $M_{l,k}$ was defined as an alternating sum of the symmetric expression

$$\sum_{I \subseteq \{1,2,\dots,k\}, |I|=j} \mu\left(\bigcap_{i \in I} X_i\right) = \sum_{i=j}^k \binom{i}{j} \mu(U_i),$$

where we see that each part of the Venn diagram for $X_1, \dots, X_k$ within U_i contributes to $\binom{i}{j}$ intersections of j systems. Exchanging the order of summation, the coefficient of a specific $\mu(U_i)$ evaluates to

$$\sum_{j=l}^k (-1)^{j-l} \binom{i}{j} = \sum_{j=l}^k (-1)^{j-l} \left[\binom{i-1}{j-1} + \binom{i-1}{j} \right] = \binom{i-1}{l-1},$$

and so $M_{l,k}$ is the sum of $\binom{i-1}{l-1} \mu(U_i)$, where $l \leq i \leq k$. □

Remark. Since $M_{2,k} = \sum_{i=2}^k (i-1) \mu(U_i)$ and $\sum_{i=1}^k i \mu(U_i)$ is the sum of $\mu(X_1), \dots, \mu(X_k)$, which is fixed, $M_{2,k}$ is minimal exactly when $\sum_{i=1}^k \mu(U_i) = 1 - \mu(U_0)$ is maximal. However, by repeated application of Theorem 1.1.1, it is clear that $\mu(U_k)$ is minimal for pairwise independent $X_1, \dots, X_k$, and as the reverse of Q_n is Q_n , this can also be said of $\mu(U_0)$. Thus, Theorem 1.1.9 in the case $l = 2$, as well as $l = k$, is a corollary of Theorem 1.1.1.

One distinguishing feature of Gladkov's inequality is that it can disprove the possibility of upward closed systems X_1, X_2, X_3 with $U_2 = \emptyset$ and

$$\mu(X_1 \cap X_2 \cap X_3), \mu(X_1 \cap X_2^c \cap X_3^c), \mu(X_1^c \cap X_2 \cap X_3^c), \mu(X_1^c \cap X_2^c \cap X_3) \rightarrow \frac{1}{6},$$

meaning the values cannot be arbitrarily close to the limit [6]. Although the limit case happens to be an equality case of Theorem 1.1.1 [7], e.g. when applied to X_1 and $X_2 \cup X_3$, Corollary 1.1.2 does not apply with arbitrary values approximating the limit.

Conjecture 1.5.2. *There is a Venn diagram of densities for $X_1, \dots, X_k$ that satisfies Theorems 1.1.1 and 1.1.7, but for which Theorem 1.1.9 does not hold.*

Next, we state and prove the abstract condition under which a linear inequality like Theorem 1.1.1 can be proven for k upward closed systems $X_1, \dots, X_k \subseteq Q_n$ by induction on n .

Lemma 1.5.3. *Let F be a linear function in 2^k variables with no constant term. For upward closed systems $X_1, \dots, X_k \subseteq Q_n$, we let F evaluate the 2^k -tuple $\mathbf{m}(X_1, \dots, X_k)$ whose components are $\mu(Z_1 \cap \dots \cap Z_k)$, where $Z_i = X_i$ or $Z_i = X_i^c$ for $1 \leq i \leq k$.*

For pairwise independent X_i , this tuple $\mathbf{m}$ only depends on the densities $\mu(X_i) = x_i$, so $F(\mathbf{m})$ evaluates to a function $G(x_1, \dots, x_k)$. Suppose that if $0 \leq x_i^- \leq x_i^+ \leq 1$ and $x_i = \frac{x_i^- + x_i^+}{2}$ for all i ,

$$G(x_1, \dots, x_k) \leq \frac{1}{2}G(x_1^-, \dots, x_k^-) + \frac{1}{2}G(x_1^+, \dots, x_k^+).$$

Then for all upward closed systems $X_1, \dots, X_k \subseteq Q_n$,

$$F(\mathbf{m}(X_1, \dots, X_k)) \geq G(x_1, \dots, x_k).$$

Proof. We follow Kleitman's proof of Theorem 1.1.1 [2], inducting on n . For $n = 0$, $Q_n = \{\emptyset\}$, and so all X_i are either $\emptyset$ or Q_0 . Since any two such systems are independent, equality holds. Inductively, suppose $F(\mathbf{m}) \geq G(x_1, \dots, x_k)$ is true in Q_n , and for any $X_i \subseteq Q_{n+1}$, partition $X_i = X_i^+ \sqcup X_i^-$ according to whether n is an element or not, respectively. Then X_i^+ and X_i^- correspond to upward closed systems in Q_n , whose densities are x_i^+ and x_i^- . Now $x_i = \frac{x_i^+ + x_i^-}{2}$, $x_i^- \leq x_i^+$, and since F is linear with no constant,

$$F(\mathbf{m}(X_1, \dots, X_k)) = \frac{1}{2}F(\mathbf{m}(X_1^+, \dots, X_k^+)) + \frac{1}{2}F(\mathbf{m}(X_1^-, \dots, X_k^-)).$$

The inductive hypothesis implies this is at least

$$\frac{1}{2}G(x_1^-, \dots, x_k^-) + \frac{1}{2}G(x_1^+, \dots, x_k^+),$$

which we suppose to be at least $G(x_1, \dots, x_k)$. Hence, the induction is complete. $\square$

Let $1 \leq l \leq k$. By Lemma 1.5.1, we can define a suitable F such that $F(\mathbf{m}(X_1, \dots, X_k)) = M_{l,k}$. If the X_i are pairwise independent with densities $\mu(X_i) = x_i$, then $\mu(\bigcap_{i \in I} X_i) = \prod_{i \in I} x_i$, so F evaluates to

$$G_l(x_1, \dots, x_k) = \sum_{j=l}^k (-1)^{j-l} \sum_{I \subseteq \{1, 2, \dots, k\}, |I|=j} \left(\prod_{i \in I} x_i \right).$$

Thus, $G_l(x_1, \dots, x_k)$ is a polynomial. We notice that G_l is linear in x_k , which is multiplied by the coefficient $G_{l-1}(x_1, \dots, x_{k-1})$ if $l \geq 2$, because a term involving x_k arises from its product with at least $l - 1$ other variables x_i .

Proposition 1.5.4. *If $2 \leq l \leq k$, then $G_l(x_1, \dots, x_k)$ meets the condition in Lemma 1.5.3.*

Proof. We first show that if $1 \leq l \leq k$, for any fixed value of $x_1, \dots, x_{k-1}$, $G_l(x_1, \dots, x_k)$ is increasing in x_k . For $l = 1$, this holds because in terms of $X_1, \dots, X_k$, the sum in question is $\sum_{i=1}^k \mu(U_i) = 1 - \mu(U_0)$, whence

$$G_1(x_1, \dots, x_k) = 1 - (1 - x_1) \dots (1 - x_k),$$

which increases in x_k . For $l \geq 2$, we induct on l , taking for granted that $G_{l-1}(x_1, \dots, x_{k-1})$ is increasing in x_{k-1} , and by symmetry, in any variable. This property, applied $l-1$ times, implies that $G_{l-1}(x_1, \dots, x_{k-1})$, the coefficient of x_k in $G_l(x_1, \dots, x_{k-1}, x_k)$, is at least $G_{l-1}(0, \dots, 0) = 0$. Hence, $G_l(x_1, \dots, x_k)$ is increasing in x_k , by induction.

For $2 \leq l \leq k$, let $0 \leq x_i^- \leq x_i^+ \leq 1$ and $x_i = \frac{x_i^- + x_i^+}{2}$. Then

$$\frac{1}{2}G_l(x_1^-, \dots, x_k^-) + \frac{1}{2}G_l(x_1^+, \dots, x_k^+) \geq \frac{1}{2}G_l(x_1^-, \dots, x_{k-1}^-, x_k) + \frac{1}{2}G_l(x_1^+, \dots, x_{k-1}^+, x_k),$$

seeing that the coefficient of x_k^+ , $G_{l-1}(x_1^+, \dots, x_{k-1}^+)$, is at least as large as the coefficient of x_k^- , $G_{l-1}(x_1^-, \dots, x_{k-1}^-)$, by applying the increasing property $l-1$ times. Upon replacing x_k^+ with x_k and x_k^- with x_k , they are decreased and increased by $\frac{x_k^+ - x_k^-}{2}$, respectively. We conclude that

$$\frac{1}{2}G_l(x_1^-, \dots, x_k^-) + \frac{1}{2}G_l(x_1^+, \dots, x_k^+) \geq \frac{1}{2}G_l(x_1, \dots, x_{k-1}, x_k) + \frac{1}{2}G_l(x_1, \dots, x_{k-1}, x_k)$$

after k replacement steps, which is the required condition. □

Theorem 1.1.9 follows immediately from Lemma 1.5.3 and Proposition 1.5.4.

Chapter 2

The Width of Hamming Balls

2.1 Introduction

Let P be a set and $<$ be an irreflexive transitive relation between pairs $x, y \in P$. We say that $(P, <)$ is a *poset*, short for partially ordered set, which we denote as P when the relation is implicitly known. A poset P is finite if the set P is finite. We let $x \leq y$ be the relation $x < y$ or $x = y$.

For $x, y \in P$, we say that y *covers* x if $x < y$ and there is no $z \in P$ such that $x < z$ and $z < y$. If $y \in P$ and P is finite, a maximal x with $x < y$ exists, and for this x , y covers x .

Let $C \subseteq P$ and $A \subseteq P$. If x and y are comparable for all $x \neq y$ in C , i.e. $x < y$ or $y < x$, we say that C is a *chain*. If there is no $x \neq y$ in A that is comparable, we say that A is an *antichain*.

Definition 2.1.1. *The size of the largest antichain of P is the width of P , denoted $\text{width}(P)$.*

Let $(P, <)$ be a poset such that every non-empty subset of P has a minimal element. In this case, any chain $C \subseteq P$ is well-ordered, and for $y \in P$, we define $r(y)$ to be the supremum of the order-type of all chains contained in $\{x \in P \mid x < y\}$. For a finite poset P , $r(y)$ is the maximum length of a chain in $\{x \in P \mid x < y\}$, and if $y < y'$, then $r(y) + 1 \leq r(y')$.

Let $P_k = \{x \in P \mid r(x) = k\}$ be the k -th *layer* of a poset P . For instance, in $(Q_n, \subset)$, where Q_n is the power set of $[n] = \{1, 2, \dots, n\}$, the layer $(Q_n)_k$ is the family of k -element subsets, because a set covers another set if and only if it has one more element. A layer is an antichain, so for any finite poset P , $\text{width}(P) \geq \max |P_k|$.

Definition 2.1.2. *A finite poset P is a *Sperner poset* if $\text{width}(P) = \max |P_k|$.*

While a theorem of Dilworth [12] informs us that every poset P can be partitioned into $\text{width}(P)$ many chains, finding these chains explicitly is another matter. If we did find c chains that together contain every element of P , then $\text{width}(P) \leq c$, because a chain meets an antichain in at most one element. More generally, if $m \cdot c$ chains contain every element of P with multiplicity at least m , then $m \cdot \text{width}(P) \leq m \cdot c$, so $\text{width}(P) \leq c$. On the other hand, if there is an antichain of size c , then $\text{width}(P) \geq c$. Thus, we can determine the width of a poset, such as Q_n .

Theorem 2.1.3 (Sperner [13]). Q_n is a Sperner poset and its width is $\binom{n}{\lfloor n/2 \rfloor}$.

Greene and Kleitman [14] provide an ingenious example of $|Q_n| = \sum_{k=0}^n \binom{n}{k}$ chains that partition Q_n . They encode a subset of $[n]$ with a string of n symbols, (or), with (in the position of non-elements and) in the position of elements. If) is preceded immediately by (, we suppress the pair, and we do this repeatedly until $n - 2m$ symbols are left, (only after). The chains are sets of subsets for which the $2m$ suppressed symbols are the same symbols in the same positions.

The proof strategy of Lubell [15], however, is to strengthen Sperner's result [13] by quantifying the size of antichains in P relative to the size of each layer P_k .

Definition 2.1.4. Let $\mathcal{C}$ be a multiset of chains in a finite poset P . Suppose that every chain in $\mathcal{C}$ passes through every nonempty layer, containing each element of any given layer equally often. Then we say that $\mathcal{C}$ is a unit flow.

The set of all chains that pass through every layer in P is a unit flow if and only if for all k and $y \in P_{k+1}$, the number of $x \in P_k$ such that $x < y$ does not depend on y . This is clear by counting how many chains that end in y pass through every previous layer. In particular, Q_n has a unit flow.

A necessary and sufficient condition for there to be a unit flow in P is the following.

Lemma 2.1.5 (Kleitman [18]). For all finite posets P , there is a unit flow of chains in P if and only if the following inequality holds for every antichain $A \subseteq P$:

$$\sum_{x \in A} \frac{1}{|P_{r(x)}|} \leq 1.$$

This is known as the LYM inequality [16], although it was Yamamoto [17] who recognised it first for $P = Q_n$. If true, it implies that $\frac{|A|}{\max_k |P_k|} \leq 1$ for every antichain $A \subseteq P$, so P is a Sperner poset. Kleitman proved that conversely, the LYM inequality implies the existence of a unit flow. Accordingly, posets that satisfy the property of Lemma 2.1.5 are known as KLYM posets [18] or strong Sperner posets [19]. The potential of KLYM posets was realised by Harper [20], who discovered a condition under which new KLYM posets can be obtained from old ones.

Definition 2.1.6. Let P and Q be posets. Then the set $P \times Q$ with partial order $(x_P, x_Q) \leq (y_P, y_Q)$ if $x_P \leq y_P$ and $x_Q \leq y_Q$ is called the product poset $P \times Q$.

Definition 2.1.7. Let $(P, <)$ be a finite poset, and let P_k have size p_k . If the fractions $\frac{p_{k+1}}{p_k}$, when defined, are decreasing in k , we say that P is log-concave.

Theorem 2.1.8 (Harper [20]). Let P and Q be finite posets. If both P and Q are log-concave KLYM posets, then so is their product poset $P \times Q$.

For example, the divisor lattice of $N = 2^{\alpha_1} \cdot 3^{\alpha_2} \cdot \dots \cdot \pi_n^{\alpha_n}$, where π_n is the n -th prime number, has the same structure as the product of n chains of respective sizes $\alpha_1 + 1, \dots, \alpha_n + 1$. Every chain is log-concave and KLYM, so the divisor lattice of N is KLYM. Furthermore, if P is a log-concave KLYM poset and $a, a + d, a + 2d, \dots, a + (l - 1)d$ is an arithmetic progression, then the subposet $Q = \bigcup_{i=0}^{l-1} P_{a+id}$ formed by the layers indexed by it is log-concave and KLYM [21].

Our main contribution is a straightforward generalisation of Theorem 2.1.8 that allows us to deduce that Hamming balls in Q_n are KLYM posets.

Theorem 2.1.9. Let P be a poset with a partition into antichains $\{P_{i,j}\}_{i,j \geq 0}$. It is presumed that if $P_{i,j} \neq \emptyset$ and $P_{i-1,j+1} \neq \emptyset$, then $P_{i,j+1} \neq \emptyset$, and that if $P_{i,j+1} \neq \emptyset$, then $P_{i,j} \neq \emptyset$ or $P_{i-1,j+1} \neq \emptyset$. Suppose that for all $i, j \geq 0$, $r(x) = i + j$ if $x \in P_{i,j}$, and a unit flow exists on $P_{i,j} \sqcup P_{i+1,j} \sqcup P_{i,j+1}$. Then provided the inequalities

$$\frac{|P_{i+1,j}|}{|P_{i,j}|} \geq \frac{|P_{i+2,j-1}|}{|P_{i+1,j-1}|} \quad \text{and} \quad \frac{|P_{i,j+1}|}{|P_{i,j}|} \geq \frac{|P_{i-1,j+2}|}{|P_{i-1,j+1}|}$$

hold (whenever the fractions are defined), there is a unit flow in P .

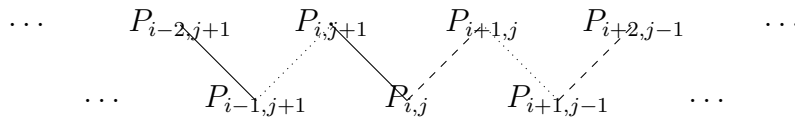


Figure 2.1: Visualisation of the relevant ratios in Theorem 2.1.9

Definition 2.1.10. Let r, n_1, n' , and $n = n_1 + n'$ be non-negative integers. The Hamming ball

$$B_r(n_1, n') = \{S \subseteq [n] : |S \Delta [n_1]| \leq r\}$$

contains the sets in Q_n which differ from $[n_1]$ by at most r elements.

Corollary 2.1.11. $B_r(n_1, n')$ is a KLYM poset, hence a Sperner poset.

2.2 The KLYM Equivalence

For the purposes of establishing Theorem 2.1.9, it will be beneficial to witness how a flow on a more intricate structure can be constructed from simpler flows, layer by layer.

Proposition 2.2.1. *Let P be a finite poset with nonempty layers $P_0, P_1, \dots, P_n$. There is a unit flow in P if and only if there is a unit flow in the subposet $P_k \sqcup P_{k+1}$ for all $0 \leq k < n$.*

Proof. Let $\mathcal{C}$ be a unit flow of P and $0 \leq k < n$. Then the chains in $\mathcal{C}$ intersect $P_k \sqcup P_{k+1}$ in pairs $\{x < y\}$. Among these pairs, every $x \in P_k$ occurs equally often, by definition of a unit flow, and similarly for $y \in P_{k+1}$. Hence, these pairs form a unit flow of $P_k \sqcup P_{k+1}$.

Conversely, let $\mathcal{C}_k$ be a unit flow of $P_k \sqcup P_{k+1}$ for $0 \leq k < n$. For all $0 \leq k \leq n$, we show by induction that a unit flow $\mathcal{F}_k$ on $P_0 \sqcup P_1 \sqcup \dots \sqcup P_k$ exists, whence there is a unit flow on P .

For $k = 0$, we let $\mathcal{F}_0 = \{\{x\} | x \in P_0\}$. If $0 \leq k < n$, we consider the multiset

$$\mathcal{F}_{k+1} = \{C \cup \{y\} | C \in \mathcal{F}_k, \{x < y\} \in \mathcal{C}_k, x \in C\}.$$

For any $x \in P_k$, there are $\frac{|\mathcal{C}_k|}{|P_k|}$ many pairs in $\mathcal{C}_k$ that involve x , and so there are $\frac{|\mathcal{C}_k|}{|P_k|}$ times as many chains in $\mathcal{F}_{k+1}$ that contain any element of $P_0 \sqcup P_1 \sqcup \dots \sqcup P_k$ as there are in $\mathcal{F}_k$. Moreover, there are $\frac{|\mathcal{C}_k|}{|P_{k+1}|} \cdot \frac{|\mathcal{F}_k|}{|P_k|}$ many chains in $\mathcal{F}_{k+1}$ that contain any $y \in P_{k+1}$. Thus, $\mathcal{F}_{k+1}$ is a unit flow, and the induction is complete. $\square$

The key lemma that guarantees a flow between two layers of a poset if a certain condition is met can be regarded as a weighted version of Hall's theorem [11], [20]. For sake of completeness, we prove it via the Ford-Fulkerson algorithm, with augmenting paths [24].

Lemma 2.2.2. *Let G be a bipartite graph with vertex set $U \sqcup V$ and edges from U to V . For $w : U \sqcup V \rightarrow \mathbb{Q} \cap [0, 1]$, denote $w(S) = \sum_{v \in S} w(v)$ for $S \subseteq U \sqcup V$, and let $w(U) = w(V) = 1$.*

Given $X \subseteq U$, let $\Gamma(X) = \{v \in V | \exists u \in X : \{u, v\} \in E(G)\}$. Suppose the weighted Hall condition holds, i.e. for all $X \subseteq U$, $w(X) \leq w(\Gamma(X))$. Then there is a multiset of edges that contains any $u \in U$ a proportion of $w(u)$ times and any $v \in V$ a proportion of $w(v)$ times.

Proof. Let N be a positive integer such that $N \cdot w(u) \in \mathbb{Z}$ for all $u \in U$ and $N \cdot w(v) \in \mathbb{Z}$ for all $v \in V$. We claim that for all $0 \leq t \leq N$, there is a multiset of t edges that contains $u \in U$ at most $Nw(u)$ times and $v \in V$ at most $Nw(v)$ times. For $t = 0$, we initialise $\emptyset$, and we prove the claim for $t + 1$ if it is true for t , whence it follows by induction.

Consider a multiset of $t < N$ edges that satisfy this property, so that some $u \in U$ appears in fewer than $Nw(u)$ edges. An *augmenting path* is a path $uv_1u_1 \dots v_iu_i$ or a path $uv_1u_1 \dots v_iu_iv$ in G such that every other edge, i.e. $\{u_1, v_1\}, \dots, \{u_i, v_i\}$, is in the multiset. Suppose that $v \in V$ is in fewer than $Nw(v)$ edges in the multiset and $uv_1u_1 \dots v_iu_iv$ is an augmenting path. In this case, we swap $\{u_1, v_1\}, \dots, \{u_i, v_i\}$ to $\{u, v_1\}, \{u_1, v_2\}, \dots, \{u_i, v\}$, obtaining a valid multiset of $t + 1$ edges.

Let $X \subseteq U$ be the set of vertices in U that occur in an augmenting path from u . Then the set of possible v in an augmenting path $uv_1u_1 \dots v_iu_iv$ from u is $\Gamma(X)$, because if $uv_1u_1 \dots v_iu_i$ is an augmenting path, v can be any neighbour of u_i , or else it occurred in the path before u_i .

The number of multiset edges between X and $\Gamma(X)$ is less than $N \cdot w(X)$, because at each $x \in X$, there are at most $Nw(x)$ multiset edges, and $u \in X$. By the Hall condition, $w(X) \leq w(\Gamma(X))$. Now if there is a $v \in \Gamma(X)$ incident with fewer than $N \cdot w(\Gamma(v))$ multiset edges, then we can make the intended swap. Otherwise, there are $N \cdot w(\Gamma(X)) \geq N \cdot w(X)$ edges in the multiset incident with $\Gamma(X)$, and so there is an edge $\{v, u'\}$ in the multiset with $v \in \Gamma(X)$ and $u' \in U \setminus X$. This is a contradiction, because there would be an augmenting path $uv_1u_1 \dots u_ivu'$.

This completes our induction. With N edges, the sum of the bounds that at most $Nw(u)$ edges are at $u \in U$ is $N \leq N$, so equality holds, and similarly, there are $Nw(v)$ edges at each $v \in V$. $\square$

This concludes the verification of the preliminary observations necessary to prove Lemma 2.1.5.

Proof of Lemma 2.1.5. Suppose that P admits a unit flow $\mathcal{C}$, and let A be an antichain. For any chain C , then, $|A \cap C| \leq 1$. If $1_C(x)$ is the indicator function of C , taking value 1 if $x \in C$ and 0 if $x \in P \setminus C$, this means $\sum_{x \in A} 1_C(x) \leq 1$. By averaging this estimate over all $C \in \mathcal{C}$, we obtain the LYM inequality, because $x \in P_k$ occurs in $C \in \mathcal{C}$ a proportion of $\frac{1}{|P_k|}$ times.

Conversely, suppose that P satisfies the LYM inequality. If $A \subseteq P_k \cup P_{k+1}$ is an antichain, then

$$\frac{|A \cap P_k|}{|P_k|} + \frac{|A \cap P_{k+1}|}{|P_{k+1}|} \leq 1, \quad \frac{|A \cap P_k|}{|P_k|} \leq \frac{|P_{k+1} \setminus A|}{|P_{k+1}|}.$$

Let $U = P_k$, $V = P_{k+1}$, and let w be the uniform distribution on U and on V . Let G be the graph whose edges are the chains $\{u, v\}$, where $u \in U$, $v \in V$. By Lemma 2.2.2, there is a unit flow on $P_k \sqcup P_{k+1}$ if the weighted Hall condition holds, i.e. $\frac{|X|}{|P_k|} \leq \frac{|\Gamma(X)|}{|P_{k+1}|}$ for all $X \subseteq P_{k+1}$. As this rearranges to the LYM inequality for the antichain $X \cup P_{k+1} \setminus \Gamma(X)$, there is a unit flow on $P_k \sqcup P_{k+1}$. Thus, by Proposition 2.2.1, there is a unit flow on P . $\square$

2.3 Product Posets and Theorem 2.1.9

To begin with, following [20], we deduce Theorem 2.1.8 from Proposition 2.2.1 and Lemma 2.2.2. We point out that Theorem 2.1.8 is in fact a special case of Theorem 2.1.9, but since the original proof [20] makes an algebra error and omits a proof that $P \times Q$ is log-concave, we take special care to complete that sketch beforehand.

Proof of Theorem 2.1.8. If $(x, y) \in P_k \times Q_l$, then $r((x, y)) \geq k + l$, because there is a length k antichain $x_0 < x_1 < \dots < x_{k-1}$ before $x_k = x$ in P and a length l antichain $y_0 < y_1 < \dots < y_{l-1}$ before $y_l = y$ in Q , so there is a length $k + l$ antichain

$$(x_0, y_0) < (x_1, y_0) < \dots < (x_k, y_0) < (x_k, y_1) < \dots < (x_k, y_{l-1})$$

in $P \times Q$ before (x, y) . However, if $(x', y') < (x, y)$ in $P \times Q$, then $x' < x$ or $y' < y$, so $r(x') < r(x)$ or $r(y') < r(y)$, and it follows by induction on $k + l$ that $r((x, y)) \leq k + l$. Therefore,

$$(P \times Q)_m = \bigcup_{k+l=m} P_k \times Q_l.$$

By Lemma 2.1.5 and Proposition 2.2.1, there is a unit flow from P_k to P_{k+1} , hence from $P_k \times Q_l$ to $P_{k+1} \times Q_l$, and similarly to $P_k \times Q_{l+1}$.

Consider the bipartite graph G on the vertex set U given by the nonempty sets $P_k \times Q_l$ with $k + l = m$ and the vertex set V given by the nonempty sets $P_k \times Q_l$ with $k + l = m + 1$, with edges from the vertex $P_k \times Q_l$ to $P_{k+1} \times Q_l$ and $P_k \times Q_{l+1}$. Let $w(P_k \times Q_l) = \frac{|P_k \times Q_l|}{|(P \times Q)_m|}$ for $P_k \times Q_l \in U$ and $w(P_k \times Q_l) = \frac{|P_k \times Q_l|}{|(P \times Q)_{m+1}|}$ for $P_k \times Q_l \in V$. Then by Lemma 2.2.2, there is a multiset of edges in G that includes $P_k \times Q_l$ in proportion to the size of that set in the respective layer, provided that for all $X \subseteq U$, $w(X) \leq w(\Gamma(X))$.

Since $U = (P \times Q)_m$ consists of the nonempty sets in the list $P_0 \times Q_m, P_1 \times Q_{m-1}, \dots, P_m \times Q_0$ and $\Gamma(\{P_{k+1} \times Q_{l-1}\}) \cap \Gamma(\{P_{k-1} \times Q_{l+1}\}) = \emptyset$, it suffices to check that $w(X) \leq w(\Gamma(X))$ for

$$\begin{aligned} X &= \{P_k \times Q_l, P_{k-1} \times Q_{l+1}, \dots, P_{k-t+1} \times Q_{l+t-1}\} \\ \Gamma(X) &= \{P_{k+1} \times Q_l, P_k \times Q_{l+1}, \dots, P_{k+1-t} \times Q_{l+t}\} \end{aligned}$$

where $k + l = m$ and $t \geq 1$. Recalling the values of w , our desired condition is

$$\frac{\sum_{i=0}^{t-1} |P_{k-i} \times Q_{l+i}|}{|(P \times Q)_{k+l}|} \leq \frac{\sum_{i=0}^t |P_{k+1-i} \times Q_{l+i}|}{|(P \times Q)_{k+l+1}|}.$$

If this condition holds, as well as the log-concave property of $P \times Q$, then we are done, because a unit flow from $(P \times Q)_m$ to $(P \times Q)_{m+1}$ occurs if we include the unit flow from $P_k \times Q_l$ to $P_{k+1} \times Q_l$ or $P_k \times Q_{l+1}$ as many times as the corresponding edge in G occurs in our multiset. In this way, every element of $P_k \times Q_l$ appears in equally many chains, and for different values of k and l with $k + l = m$, the number of chains at each element of $P_k \times Q_l$ is equal. By Proposition 2.2.1, these unit flows between consecutive layers of $P \times Q$ combine into a unit flow on $P \times Q$.

To establish the required inequalities, let $|P_k| = p_k$ and $|Q_l| = q_l$.

Multiplying either side of the unit flow condition by $|(P \times Q)_m| \cdot |(P \times Q)_{m+1}|$, it reads

$$\begin{aligned} \left(\sum_{i=0}^{t-1} p_{k-i} q_{l+i} \right) \left(\sum_j p_{k+1-j} q_{l+j} \right) &\leq \left(\sum_{i=0}^t p_{k+1-i} q_{l+i} \right) \left(\sum_j p_{k-j} q_{l+j} \right), \\ \sum_{i=0}^{t-1} \sum_j p_{k-i} p_{k+1-j} q_{l+i} q_{l+j} &\leq \sum_{i=0}^t \sum_j p_{k+1-i} p_{k-j} q_{l+i} q_{l+j}. \end{aligned}$$

Notice that if $j < 0$, the inequality is linear in q_{l+j} , with coefficients $\sum_{i=0}^{t-1} p_{k-i} p_{k+1-j} q_{l+i}$ and $\sum_{i=0}^t \sum_j p_{k+1-i} p_{k-j} q_{l+i}$. By log-concavity of P , $p_{k-i} p_{k+1-j} \leq p_{k+1-i} p_{k-j}$, because $j < 0 \leq i$.

Also notice that if $j > t$, the inequality is linear in p_{k+1-j} , with coefficients $\sum_{i=0}^{t-1} p_{k-i} q_{l+i} q_{l+j}$ and $\sum_{i=0}^t p_{k+1-i} q_{l+i} q_{l+j-1} = \sum_{i=-1}^{t-1} p_{k-i} q_{l+i+1} q_{l+j-1}$. By log-concavity of Q , $q_{l+i} q_{l+j} \leq q_{l+i+1} q_{l+j-1}$.

According to Harper [20], we are to estimate in this way not for $j < 0$ and $j > t$, but rather whenever $j \leq i$ and $j > i$, respectively. Although $t - 1$ duplicate terms would emerge if we did so, with our approach, there is an equality between the terms that were not compared so far.

$$\left(\sum_{i=0}^{t-1} p_{k-i} q_{l+i} \right) \left(\sum_{j=0}^t p_{k+1-j} q_{l+j} \right) = \left(\sum_{i=0}^t p_{k+1-i} q_{l+i} \right) \left(\sum_{j=0}^{t-1} p_{k-j} q_{l+j} \right)$$

Finally, to show that $P \times Q$ is log-concave, we quote Canfield [22], [23].

$$\begin{aligned} |(P \times Q)_k|^2 &\geq |(P \times Q)_{k+1}| \cdot |(P \times Q)_{k-1}| \\ \left(\sum_i p_{k-i} q_i \right) \cdot \left(\sum_j p_{k-j} q_j \right) &\geq \left(\sum_i p_{k+1-i} q_i \right) \cdot \left(\sum_j p_{k-1-j} q_j \right) \\ \sum_{i,j} (p_{k-i} p_{k-j} - p_{k+1-i} p_{k-1-j}) (q_i q_j - q_{i-1} q_{j+1}) &\geq 0 \end{aligned}$$

By log-convexity of P and Q , these two brackets are both ≥ 0 if $i \leq j$ and both ≤ 0 if $i \geq j$. $\square$

While the log-concave property in Theorem 2.1.8 may seem unmotivated, it is in fact natural. Indeed, considering the structure of $B_r(n_1, n')$, which resembles a product poset, a similar property allows us to deduce that a unit flow exists. Hence, we prove Corollary 2.1.11 of Theorem 2.1.9.

Proof of Corollary 2.1.11. Let $P = B_r(n_1, n')$ and let $P_{i,j} \subseteq P$ contain the subsets of $[n]$ that arise by deleting $\min\{n_1, r\} - i$ elements from $[n_1]$ and including j elements from $[n_1 + n'] \setminus [n_1]$. This way, $P_{0,0}$ contains the minimal elements of P , with as many elements deleted from $[n_1]$ as possible. Further, there is a set in $P_{i,j}$ exactly if $i \leq \min\{n_1, r\}$, $j \leq n'$, and $\min\{n_1, r\} - i + j \leq r$, in which case

$$|P_{i,j}| = \binom{n_1}{\min\{n_1, r\} - i} \binom{n'}{j}.$$

The log-concavity of Q_n implies that $\frac{|P_{i+1,j}|}{|P_{i,j}|}$ decreases in i and $\frac{|P_{i,j+1}|}{|P_{i,j}|}$ decreases in j .

Let $x \in P_{i,j}$. Since returning deleted elements before including elements keeps a set within P , $r(x) \geq i + j$. On the other hand, by keeping track of set sizes, $r(x) \leq i + j$. Hence, $r(x) = i + j$.

By symmetry of permuting elements of $[n_1]$ and $[n_1 + n'] \setminus [n_1]$, there is a unit flow from $P_{i,j}$ to $P_{i+1,j}$ and from $P_{i,j}$ to $P_{i,j+1}$, provided they are nonempty.

Thus, the conditions of Theorem 2.1.9 are met by the Hamming ball $B_r(m, l)$. $\square$

To conclude, we prove Theorem 2.1.9, our generalisation of Theorem 2.1.8.

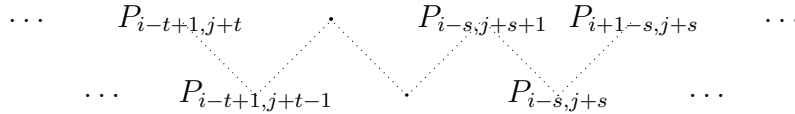


Figure 2.2: Parts of P compared in the weighted Hall condition

Proof of Theorem 2.1.9. By Proposition 2.2.1, it suffices to guarantee a unit flow from layer $P_k = \bigcup_{i+j=k} P_{i,j}$ to layer $P_{k+1} = \bigcup_{i+j=k+1} P_{i,j}$ for each k . Since there is a unit flow from $P_{i,j}$ to $P_{i+1,j}$ and from $P_{i,j}$ to $P_{i,j+1}$, these can be combined into a unit flow from P_k to P_{k+1} if the weighted Hall condition

$$\frac{\sum_{l=s}^{t-1} |P_{i-l, j+l}|}{\sum_l |P_{i-l, j+l}|} \leq \frac{\sum_{l=s}^t |P_{i+1-l, j+l}|}{\sum_l |P_{i+1-l, j+l}|}$$

holds for all $s < t$, by applying Lemma 2.2.2 in the very same way as in proving Theorem 2.1.8.

To check this condition, rather than expanding, we rearrange to

$$\frac{\sum_l |P_{i+1-l, j+l}|}{\sum_l |P_{i-l, j+l}|} \leq \frac{\sum_{l=s}^t |P_{i+1-l, j+l}|}{\sum_{l=s}^{t-1} |P_{i-l, j+l}|}$$

and show that by incrementing t or decrementing s , the right-hand side cannot increase.

Now if $P_{i-t,j+t}$ is non-empty, then given that $\frac{|P_{i-l,j+l+1}|}{|P_{i-l,j+l}|}$ decreases with l , $\frac{|P_{i-l,j+l+1}|}{|P_{i-l,j+l}|} \geq \frac{|P_{i-t,j+t+1}|}{|P_{i-t,j+t}|}$ holds for $s \leq l \leq t-1$. Hence,

$$\frac{\sum_{l=s}^{t-1} |P_{i-l,j+l+1}|}{\sum_{l=s}^{t-1} |P_{i-l,j+l}|} \geq \frac{|P_{i-t,j+t+1}|}{|P_{i-t,j+t}|},$$

where $\sum_{l=s}^t |P_{i+1-l,j+l}| \geq \sum_{l=s}^{t-1} |P_{i-l,j+l+1}|$. By estimating fractions, it follows that

$$\frac{\sum_{l=s}^t |P_{i+1-l,j+l}|}{\sum_{l=s}^{t-1} |P_{i-l,j+l}|} \geq \frac{\sum_{l=s}^{t+1} |P_{i+1-l,j+l}|}{\sum_{l=s}^t |P_{i-l,j+l}|}.$$

For the same reason, provided that $\frac{|P_{i+1-l,j+l}|}{|P_{i-l,j+l}|}$ decreases with $-l$, if we add $|P_{i+2-s,j+s-1}|$ to the numerator and $|P_{i+1-s,j+s-1}|$ to the denominator,

$$\frac{\sum_{l=s}^t |P_{i+1-l,j+l}|}{\sum_{l=s}^{t-1} |P_{i-l,j+l}|} \geq \frac{\sum_{l=s-1}^t |P_{i+1-l,j+l}|}{\sum_{l=s-1}^{t-1} |P_{i-l,j+l}|}.$$

Thus, without loss of generality, $P_{i-t,j+t} = \emptyset$ and $P_{i+1-s,j+s-1} = \emptyset$. Observe that our presumptions guarantee that if $P_{i-l,j+l}$, $s \leq l \leq t-1$, were the nonempty parts of layer k , then the nonempty parts of layer $k+1$ would be $P_{i-l,j+1+l}$, $s+1 \leq l \leq t-1$ and possibly $P_{i-s,j+1+s}$ or $P_{i-t,j+1+t}$. Hence, by inducting on k , the values of l for which $P_{i-l,j+l} \neq \emptyset$ are in an interval, and so the case we reduced our condition to is the trivial case. $\square$

Chapter 3

Layered Subgraphs of the Hypercube

3.1 Introduction

Let n be a positive integer. The n -dimensional *hypercube graph* $\mathcal{Q}_n$, with $V(\mathcal{Q}_n) = \{0, 1\}^n$, involves an edge between $x, y \in \{0, 1\}^n$ if and only if x and y differ on exactly one coordinate. The k -th *edge-layer* $\mathcal{Q}_n^k$ is the subgraph of $\mathcal{Q}_n$ induced by the vertices containing exactly $k - 1$ or k coordinates equal to 1, where $1 \leq k \leq n$.

Let $ex(\mathcal{Q}_n, G)$ be the most edges in a subgraph of $\mathcal{Q}_n$ that has no subgraph isomorphic to G . By averaging over embeddings of $\mathcal{Q}_n$ into $\mathcal{Q}_{n+1}$, it is clear that $\frac{ex(\mathcal{Q}_n, G)}{|E(\mathcal{Q}_n)|}$ is decreasing, and its limit $\rho(G)$ is called the (edge-)Turán density of G [25]. In determining $\rho(G)$, the notions of a cubical graph and of a layered graph play an important role.

Definition 3.1.1. A graph G is *cubical* if there exists n such that $G \leq \mathcal{Q}_n$. A cubical graph G is *layered* if there exist n, k such that $G \leq \mathcal{Q}_n^k$.

If G is not layered, $\rho(G) \geq \frac{1}{2}$, by taking the disjoint union of the layers $\mathcal{Q}_n^k$ over all even k or all odd k . For instance, $\mathcal{Q}_2$, i.e. the cycle of length 4, is not layered. Erdős enquired about how large $ex(\mathcal{Q}_n, C_{2t})$ can be, where C_l is the cycle of l edges [26], offering \$100 for a proof or disproof that $\rho(C_4) = \frac{1}{2}$. The best known upper bound is $\rho(C_4) \leq 0.6032$ [27]. $E(\mathcal{Q}_n)$ can be partitioned into three sets, each of which do not contain a C_6 , and so $\rho(C_6) \geq \frac{1}{3}$ [28]. While there is no such partition into a constant number of sets for $G = C_{10}$ [29], $\rho(C_{10}) > \frac{1}{42}$ holds [30]. On the other hand, if $G \leq \mathcal{Q}_n^k$, such that $[n]$ can be coloured using k colours where for all $x \in V(G)$, the colouring on $\{i \in [n] : x_i = 1\}$ is injective, then $\rho(G) = 0$ [31]. Although this result suffices to show $\rho(C_{2t}) = 0$ if $t \geq 6$ or $t = 4$ [31], it is not a necessary condition of $\rho(G) = 0$ [32].

Recall that the length of the shortest cycle in a graph G , if it exists, is known as the girth of G . Axenovich, Martin and Winter found a characterisation of layered graphs, exhibited graphs with girth $g \in \{4, 6, 8\}$ that are cubical but not layered, and asked whether graphs of arbitrarily large girth can be layered and not cubical [33]. We shall answer this question completely.

Theorem 3.1.2. *For every even $g \geq 6$, there exists a graph of girth g that is cubical but not layered.*

With a more thorough analysis of our graphs and their metric properties, we obtain an example of a graph with girth g that are an induced subgraph of $\mathcal{Q}_n$.

Theorem 3.1.3. *For every even $g \geq 8$, for sufficiently large n , there exists an induced subgraph of $\mathcal{Q}_n$ of girth g that is not layered.*

A helpful idea for proving Theorem 3.1.2 is to consider the metric properties of embeddings. How does the graph distance in a given subgraph relate to the metric distance in the hypercube? The core of our proof is an iterative procedure (Lemma 3.3.2) that prepares a cubical tree with a special property, which can only be embedded in a layer if certain pairs of vertices are at a large distance. By suitably adding paths to this tree, we create a graph that is cubical but not layered. The proof of Theorem 3.1.3 is more involved, as we need to preserve certain additional properties during the proof and improve the tree we begin with. However, we modify this tree in the same way as in our proof of Theorem 3.1.2.

The minimum degree of our examples for Theorems 3.1.2 and 3.1.3 grows factorially in $\frac{g}{2}$, and so it is unknown whether n can be only exponential in g .

Question 3.1.4. *As $g \rightarrow \infty$, what is the growth rate of the least n for which there is a girth g non-layered subgraph of $\mathcal{Q}_n$?*

3.2 Characterisation of Cubical and Layered Graphs

The heart of our argument is a special relationship between particular embeddings of a graph G into a hypercube and certain edge-labellings of G .

Recall that an *embedding* of a graph G into a graph H is a graph isomorphism from G to a subgraph of H . Let f be an embedding of a cubical graph G in $\mathcal{Q}_n$. As a helpful descriptive device, we label edges $xy \in E(\mathcal{Q}_n)$ according to their *direction*, the coordinate where they differ.

Definition 3.2.1. *Let G be a graph and $\chi : E(G) \rightarrow [n]$ be a labelling of its edges. We say that the labelling χ arises from the embedding f if for all $xy \in E(G)$, $\chi(xy)$ is the direction of $f(x)f(y)$.*

In a walk from node x to node y in $\mathcal{Q}_n$, the coordinates of consecutive nodes differ by the label of the edge between them. By recursing along the walk, we see that y differs from x exactly at coordinates that occur an odd number of times as the label of an edge in the walk. Thus, if $x \neq y$, then some label must occur an odd number of times along a walk from x to y , but if $x = y$, then every label occurs an even number of times. We call a labelling that satisfies this key property a *cubical labelling*. Havel and Morávek [34] discovered that if G admits a cubical labelling, then it can be realised as a subgraph of $\mathcal{Q}_n$.

Lemma 3.2.2 (Havel and Morávek [34]). *Let G be a connected graph. A labelling $\chi : E(G) \rightarrow [n]$ arises from an embedding f of G in $\mathcal{Q}_n$ if and only if χ is cubical.*

Proof. It remains to verify that if $\chi : E(G) \rightarrow [n]$ is a cubical, then it arises from an embedding f .

Let $x \in V(G)$ be arbitrary, for which we let $f(x) = \mathbf{0} \in V(\mathcal{Q}_n)$. If $y \in V(G)$, there is a walk from x to y , because G is connected. In some such walk, let $f(y)$ be such that $f(y)_i$ is the number of times label $i \in [n]$ occurs, modulo 2. Since the labelling is cubical, the label i occurs an even number of times when walking from x to y and back, so $f(y)$ does not depend the walk in question. Further, if $x \neq y$, then there is a label $i \in [n]$ that occurs an odd number of times along a walk from x to y , so $f(y)_i \neq f(x)_i$. Hence, f is an embedding of G into $\mathcal{Q}_n$.

In our definition, $f(y)$ could be defined using any walk from x to y , and so if $yz \in E(G)$, we could consider the walk from x to z given by some walk from x to y , followed by yz . Thus, we see that $f(y)$ and $f(z)$ only differ only at $\chi(yz)$, and so χ arises from f . $\square$

A useful corollary of this lemma is the key idea of our proof.

Corollary 3.2.3. *Let χ be a cubical labelling of a graph G consisting of $l + 1$ vertex-disjoint P_l (paths of l edges) from x to y . Then it is impossible for some P_l to be labelled with l distinct labels.*

Proof. By Lemma 3.2.2, χ arises from an embedding f of G . If some path of length l from $f(x)$ to $f(y)$ changes l directions, then $f(x)$ and $f(y)$ differ in those coordinates. Therefore, any path of length k from $f(x)$ to $f(y)$ contains exactly those l labels. However, if the $l + 1$ edges at x use k labels, then by the pigeonhole principle, two neighbours of x coincide in f , a contradiction. $\square$

If $l \geq 3$, the graph in Corollary 3.2.3 is cubical, because if we label each P_l with $j, 1, 2, \dots, l-2, j$, where j is a label unique to the path, we obtain a cubical labelling. However, if $l = 3$, this girth 6 graph is not layered, for the labels along a P_3 are distinct in a layered labelling, as we will see.

Layered graphs can be characterised in a similar way [33]. In $\mathcal{Q}_n^k$, every edge has one endpoint with $k - 1$ 1s and another with k 1s. Suppose a walk from x to y has direction i exactly on the first edge xx' and final edge $y'y$. If $x_i = 0$, then $x'_i = 1$, $y'_i = 1$, and $y_i = 0$, so both x and y contain k 1s. If $x_i = 1$, then $x'_i = 0$, $y'_i = 0$, and $y_i = 1$, so both x and y contain k 1s. Either way, the walk has an even number of edges.

Definition 3.2.4. A cubical labelling $\chi : E(G) \rightarrow [n]$ is a layered labelling if every walk where a label occurs only on the first and last edge is of even length.

Lemma 3.2.5 (Axenovich, Martin and Winter [33]). Let G be a connected graph and $\chi : E(G) \rightarrow [n]$ be a cubical labelling. Then χ arises from an embedding f of G into $\mathcal{Q}_n^k$, for suitable k , if and only if χ is layered.

Proof. Let us show the converse.

By Lemma 3.2.2, χ arises from an embedding f of G in $\mathcal{Q}_n$. In fact, for an arbitrary $x \in V(G)$, we could have specified $f(x)$ arbitrarily. Since $\mathcal{Q}_n$ is bipartite, G is a connected bipartite graph, and so there is a unique bipartition of G into $V(G) = V_0 \sqcup V_1$, where $x \in V_0$.

For each $i \in [n]$, let us determine $f(x)_i$ as follows. If i is not a label of G , $f(x)_i = 0$. If i does occur as a label, then since G is connected, there is a walk from x that uses the label i , and let yz be the first edge in this walk that is labelled i , where y is visited before z , i.e. $f(y)_i = f(x)_i$. For some such walk, if $y \in V_0$, then let $f(x)_i = 0$, and if $y \in V_1$, then let $f(x)_i = 1$.

Specifying f in this way guarantees for *one* of the edges y_0y_1 of label i , where $y_0 \in V_0$, $y_1 \in V_1$, that $f(y_0)_i = 0$ and $f(y_1)_i = 1$. However, since χ is layered, there are an even number of edges between two consecutive instances of the label i along a walk in G . If $z'z$ is the first instance of the label i after yy' , then since vertices in V_0 and V_1 alternate along the walk, $y \in V_0$ if and only if $z \in V_0$, but also $f(y')_i = f(z')_i$. Hence, this property holds for *all* edges $y_0y_1 \in E(G)$ of label $\chi(y_0y_1) = i$.

Therefore, if $y_0y_1 \in E(G)$ and $y_0 \in V_0$, then there is one more 1 among the coordinates of $f(y_1)$ than among the coordinates of $f(y_0)$. Let $k - 1$ be the number of 1s among the coordinates of $f(x)$. By considering a walk from x to y , the number of 1s in $f(y)$ is $k - 1$ if $y \in V_0$ and k if $y \in V_1$. Thus, the image of f is a subgraph of $\mathcal{Q}_n^k$. $\square$

From Lemma 3.2.5, let us deduce a significant partial result in Theorem 3.1.2.

Theorem 3.2.6 (Axenovich, Martin and Winter [33]). *There is a girth 8 subgraph of $\mathcal{Q}_4$ that cannot embed into a layer.*

Proof. The graph $\mathcal{Q}_4$ induces a cycle C_8 on vertices

$$\{0000, 0001, 0011, 0111, 1111, 1110, 1100, 1000\}.$$

In addition, it is possible to include the two paths induced by

$$\{0000, 0100, 0101, 1101, 1111\} \quad \text{and} \quad \{0001, 1001, 1011, 1010, 1110\}.$$

Notice that the pair $\{0000, 1111\}$ is connected by three paths of length 4 with disjoint interiors. In a cubical edge-labelling, if one of these paths has 4 distinct labels, each of the paths must use these four labels. Furthermore, if the labelling is layered, by parity, only two of the four labels can occur at a given endpoint. The labels at a vertex are distinct, so this is a contradiction.

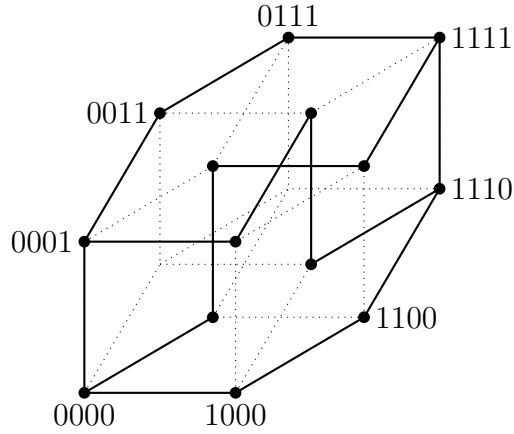


Figure 3.1: The girth 8 graph due to [33]

Therefore, in a layered labelling, a path from 0000 to 1111 includes at least one label twice. This label must be the first and last label, as there are two edges between them. These middle two edges, then, are labelled using the same two labels, i and j , and in the same order from 0000 to 1111, since the labelling is layered. The same can be said regarding the triple P_4 from 0001 to 1110, so it follows that for their middle two edges, i and j are the labels. This contradicts how edge $\{0000, 1000\}$ does not use i or j , so there cannot be layered labelling of this graph. $\square$

3.3 Distances and Labelled Walks

Recall that in a graph G , the *distance* $d_G(x, y)$ between $x, y \in V(G)$ is the number of edges in the shortest walk from x to y .

Definition 3.3.1. Let G be a graph and P be a set of unordered pairs $\{x, y\}$, where $x, y \in V(G)$. We say that (G, P) is t -distancing, where t is a positive integer, if the following hold:

- (i) For some n , there is an embedding f of G into $\mathcal{Q}_n$ such that whenever $\{x, y\} \in P$, the hypercube distance $d(f(x), f(y)) = t$;
- (ii) If ϕ is an embedding of G into a hypercube layer $\mathcal{Q}_n^k$, then for some pair $\{x, y\} \in P$, the hypercube distance $d(\phi(x), \phi(y)) = t + 2$.

Our strategy for generating high-girth graphs that are cubical and not layered involves inductively creating a tree T and a suitable P that makes (T, P) $2t$ -distancing.

Lemma 3.3.2. Let T be a tree and P be a set of pairs of vertices in T . If (T, P) is t -distancing, there exists a tree T' and a set P' of pairs of vertices in T' such that (T', P') is $(t + 2)$ -distancing.

Proof. For $s = t + 4$, let T' be the tree obtained from T in the following way. For each vertex x that participates in a pair in P , we add s new leaves $x_1, \dots, x_s$ adjacent to x . Now if the new leaves at y are $y_1, \dots, y_s$, let P_{xy} be the set of all pairs $\{x_i, y_j\}$, $1 \leq i, j \leq s$, so that $P' = \bigcup_{\{x, y\} \in P} P_{xy}$.

Let us prove that (T', P') is $(t + 2)$ -distancing. It is known that (T, P) is t -distancing, so there is an embedding f of T into $\mathcal{Q}_n$, for some n , where $d(f(x), f(y)) = t$ whenever $\{x, y\} \in P$. Using the labelling of $E(T)$ that arises from this embedding, we label the leaf edges of $E(T') \setminus E(T)$ with unique labels, distinct from those of $E(T)$. Clearly, this is a cubical labelling of T' . Thus, by Lemma 3.2.2, there is an embedding f' of T' into $\mathcal{Q}_{n+s|P|}$ that respects this labelling.

Along a path from x_i to y_j that begins with $x_i x$ and ends with yy_j , the label of $x_i x$ and of yy_j occurs only once. The hypercube distance in a cubical embedding is the number of labels that occur an odd number of times, whence

$$d(f'(x_i), f'(y_j)) = d(f'(x), f'(y)) + 2 = d(f(x), f(y)) + 2 = t + 2.$$

Therefore, T' is embedded such that whenever $\{x', y'\} \in P'$, the distance $d(f'(x'), f'(y')) = t + 2$.

Since (T, P) is t -distancing and $T \leq T'$, we also know that whenever ϕ is an embedding of T' into $\mathcal{Q}_m$, there must be an $\{x, y\} \in P$ with $d(\phi(x), \phi(y)) = t + 2$. T' was defined with s leaves $x_1, \dots, x_s$ at x and $y_1, \dots, y_s$ at y , where $s - (t + 2) = 2$. This way, in the labelling that arises from g , we can choose xx_i and yy_j to have two distinct labels other than the $t + 2$ coordinates where $\phi(x)$ and $\phi(y)$ differ, yielding $d(\phi(x_i), \phi(y_j)) = t + 4$, a pair in P' at distance $t + 4$. $\square$

3.4 Proof of Theorem 3.1.2

We begin our proof by establishing that a t -distancing tree T and pair set P exists for all t . By induction, thanks to Lemma 3.3.2, it suffices to give an example for $t = 1$ and for $t = 2$.

The path P_3 can be viewed as the path $(00, 01, 11, 10)$ in $\mathcal{Q}_2$, but in a layered embedding, the directions of its three edges are distinct. Hence, P_3 and its pair of endpoints is 1-distancing.

Lemma 3.4.1. *Let P_6 be the path on 6 edges with vertices labelled $v_0, v_1, \dots, v_6$, and let P be the pairs of vertices $\{\{v_0, v_4\}, \{v_1, v_5\}, \{v_2, v_6\}\}$. Then (P_6, P) is 2-distancing.*

Proof. To show condition (i) of Definition 3.3.1, consider the following embedding of P_6 into $\mathcal{Q}_3$:

$$v_0 \mapsto 000, \quad v_1 \mapsto 100, \quad v_2 \mapsto 110, \quad v_3 \mapsto 010,$$

$$v_4 \mapsto 011, \quad v_5 \mapsto 111, \quad v_6 \mapsto 101.$$

Clearly, $d(f(v_0), f(v_4)) = 2$, $d(f(v_1), f(v_5)) = 2$, and $d(f(v_2), f(v_6)) = 2$, as required.

To show condition (ii) of Definition 3.3.1, let χ be a layered labelling of P_6 that arises from a cubical embedding f . Then

- (a) Lemma 3.2.5 implies the same label cannot occur twice with one or three intermediary edges;
- (b) Lemma 3.2.2 implies that along any path, some label occurs an odd number of times.

Without loss of generality, let $\chi(v_0v_1) = 1$.

Using (b), we may assume that $\chi(v_1v_2) = 2$.

By (a), $\chi(v_2v_3) \neq 1$. By (b), $\chi(v_2v_3) \neq 2$. We may assume that $\chi(v_2v_3) = 3$.

If $\chi(v_3v_4)$ is not 1, 2, or 3, then $d(f(v_0), f(v_4)) = 4$.

Otherwise, like in the preceding step, $\chi(v_3v_4) = 1$.

Next, unless $d(f(v_1), f(v_5)) = 4$, we find $\chi(v_4v_5) = 2$.

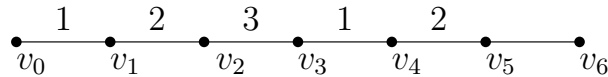


Figure 3.2: The edge-labels forced on P_6 during the proof of Lemma 3.4.1

Finally, since (b) implies $\chi(v_5v_6) \neq 3$, it must be the case that $d(f(v_2), f(v_6)) = 4$. □

Corollary 3.4.2. *For all positive integers t , there exists a tree T and a set P of pairs of vertices in T such that (T, P) is t -distancing.*

We are now ready to prove Theorem 3.1.2, which we restate below for convenience.

Theorem 3.4.3. *For every even $g \geq 6$, there exists a graph of girth g that is cubical but not layered.*

Proof. By Corollary 3.4.2, there is a tree T and a set P of pairs of vertices in T that is t -distancing. For each pair $\{x, y\} \in P$, add $t + 3$ vertex-disjoint paths of $t + 2$ edges between x and y , thus forming a graph G . Let S_{xy} be the set of paths from x to y , which we will refer to as a *spindle*.

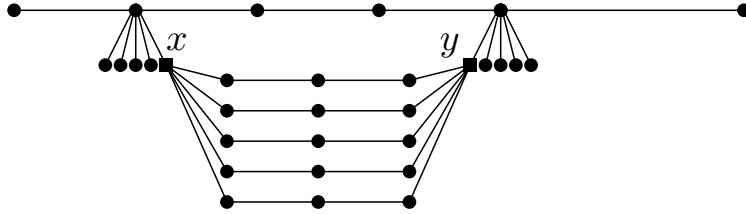


Figure 3.3: A spindle (for $t = 2$) between vertices x and y

In our examples of a t -distancing tree for $t = 1, 2$, the minimum distance in the tree between a pair in P was $t + 2$. Hence, by induction, we may suppose the minimum tree distance between a pair in P is $t + 2$. Since these pairs are now joined with paths of $t + 2$ edges, the girth of G is $2(t + 2) = g$, which ranges over all even values $g \geq 6$.

First, we show that G is cubical. To this end, we provide a cubical labelling $\chi : E(G) \rightarrow \mathbb{N}$ and apply Lemma 3.2.2. Since (T, P) is t -distancing, we begin with a labelling of T that arises from an embedding f of T into $\mathcal{Q}_n$ where pairs in P are t apart. We can extend this to a labelling of G if we label the spindle S_{xy} for each $\{x, y\} \in P$ as follows.

For $\{x, y\} \in P$, let I_{xy} be the set of coordinates that $f(x)$ and $f(y)$ differ on, so that $|I_{xy}| = d(f(x), f(y)) = t$. For every x - y path in S_{xy} , we introduce a unique new edge label, for a total of $(t + 3) \cdot |P|$ new labels. This labels the first and final edge of the path, while I_{xy} labels the central t edges in some order. Since this corresponds to embedding each $(t + 2)$ -path in a unique translate of $\mathcal{Q}_n$ within $\mathcal{Q}_{n+(t+3)|P|}$, the labelling is cubical.

Notice that G is not layered, because we know from Corollary 3.2.3 that a spindle S_{xy} cannot be embedded such that every edge in some path is in a different direction, i.e. $d(f(x), f(y)) = t + 2$. By Definition 3.3.1, in a layered embedding of T , this must be the case for some $\{x, y\} \in P$. $\square$

3.5 Proof of Theorem 3.1.3

In our embedding of the trees $T = P_3$ and $T = P_6$, there were adjacent hypercube vertices in the image that were not adjacent in the tree. Although this is an essential feature of a 1-distancing (T, P) , for $t \geq 2$, there is a t -distancing (T, P) where T embeds as an induced hypercube subgraph.

Lemma 3.5.1. *Let T be the tree on 9 vertices that is the path $v_0, v_1, \dots, v_7$ together with a vertex v_8 adjacent to v_2 . Also, let*

$$P = \{\{v_0, v_4\}, \{v_0, v_6\}, \{v_3, v_7\}, \{v_5, v_8\}\}.$$

Then (T, P) is 2-distancing. Moreover, if we form T' from T by adding 5 new leaves to each of the vertices v_4, v_6, v_7, v_8 and P' from P by replacing each pair that involves these vertices with one pair for each respective leaf, then (T', P') is 3-distancing.

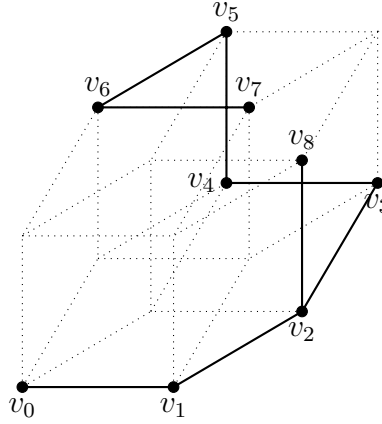


Figure 3.4: Embedding from Lemma 3.5.1

Proof. On Figure 3.4, T is an induced subgraph of Q_4 where pairs in P have distance 2. We read off that $v_0, v_1, \dots, v_8$ map respectively to 0000, 1000, 1100, 1110, 0110, 0111, 0011, 1011, 1101. If we add leaves to T in unique directions of a sufficiently large hypercube, the resulting tree will be induced. Further, when we replace a vertex of a pair in P with its leaf, the pair distance increases by 1. Hence, T' embeds as an induced subgraph of Q_{24} where pairs in P' are at distance 3.

Suppose that whenever T is embedded in a hypercube layer, there is a pair $\{x, y\} \in P$ at distance 4. In T' , there are five leaves at one of them, say at y . Similarly to Lemma 3.3.2, applying the pigeonhole principle, we see that one of these five leaves, yy' , is in a direction at which the images of x and y do not differ. Thus, the images of x and y' have distance 5.

It remains for us to show that in every layered labelling χ of T , there is a pair from P whose distance is 4 in any embedding f that arises from χ . When proving Lemma 3.4.1, we observed the following:

- (a) Lemma 3.2.5 implies the same label cannot occur twice with one or three intermediary edges;
- (b) Lemma 3.2.2 implies that along any path, some label occurs an odd number of times.

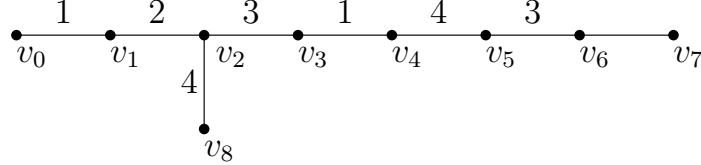


Figure 3.5: The edge-labels forced on T during the proof of Lemma 3.5.1.

Similarly, here, we may presume $\chi(v_0v_1) = 1$, $\chi(v_1v_2) = 2$, $\chi(v_2v_3) = 3$, and $\chi(v_2v_8) = 4$.

Unless $d(f(v_0), f(v_4)) = 4$, it must be the case that $\chi(v_3v_4) \in \{1, 2, 3\}$, and by (a), $\chi(v_3v_4) = 1$.

Unless $d(f(v_8), f(v_5)) = 4$, by (a), we see that $\chi(v_4v_5) = 4$.

Unless $d(f(v_0), f(v_6)) = 4$, $\chi(v_5v_6) \in \{2, 3, 4\}$, whence by (a), $\chi(v_5v_6) = 3$.

However, 1 cannot label v_6v_7 , because then, every label on the path from v_8 to v_7 would occur twice, contradicting (b). Therefore, $d(f(v_3), f(v_7)) = 4$. $\square$

Starting from the tree defined in Lemma 3.5.1, the graph construction we use to prove Theorem 3.1.3 is the same as for Theorem 3.1.2, but we require further properties.

Corollary 3.5.2. *For all $t \geq 2$, there is a tree T and a set of pairs of its vertices P such that:*

- (i) *there exist n and an embedding f of T into $\mathcal{Q}_n$ where for all $\{x, y\} \in P$, $d(f(x), f(y)) = t$,*
- (ii) *in the labelling $\chi : E(T) \rightarrow [n]$ that arises from f , every label but 1, 2, and 4 occurs once,*
- (iii) *if T embeds in a hypercube layer, some pair $\{x, y\} \in P$ embeds at hypercube distance $t + 2$,*
- (iv) *if $\{x, y\} \in P$, then the path from x to y can consist of $t + 2$ or $t + 4$ edges, the labels of which occur once or twice,*
- (v) *if $\{x, y\} \in P$ and $t \geq 3$, x or y is a leaf of T and the label of its leaf is not 1, 2, or 4,*
- (vi) *the image of f is an induced subgraph of $\mathcal{Q}_n$.*

Proof. If $t = 2$, the tree (T, P) described in the proof of Lemma 3.5.1 clearly satisfies each of these properties. When applying Lemma 3.3.2 and 3.5.1, each new leaf is embedded in a unique direction, so the claim follows by induction. $\square$

We conclude by proving Theorem 3.1.3, which we restate below for convenience.

Theorem 3.5.3. *For every even $g \geq 8$, for sufficiently large n , there exists an induced subgraph of $\mathcal{Q}_n$ of girth g that is not layered.*

Proof. Corollary 3.5.2 yields (T, P) and an embedding f of T in $\mathcal{Q}_n$. Proceeding in the same way as in proving Theorem 3.1.2, we expand T with spindles S_{xy} , formed by $t + 3$ paths of $t + 2$ edges from x to y , for every $\{x, y\} \in P$. By (iii) and Corollary 3.2.3, the resulting graph G is not layered. By (iv), the girth of G is $2(t + 2) = g$, which ranges over even $g \geq 8$.

Next, we embed G into $\mathcal{Q}_{n+(t+3)|P|}$. Let $\chi : E(T) \rightarrow [n]$ be the labelling that arises from f . If $\{x, y\} \in P$, then by (i) and (iv), along the path from x to y in T , there are t labels that occur once, the rest occurring twice. In order of appearance, let these be $i_1, i_2, \dots, i_t$. For each path in S_{xy} , let j be a new label unique to the path, and from x to y , label the path $j, i_t, i_{t-1}, \dots, i_1, j$, thus embedding it in a translate of $\mathcal{Q}_n$ in direction j . Hence, f extends to an embedding of G , which we may call f , and it remains to check that it maps non-adjacent vertices to non-adjacent vertices.

Let $w \notin V(T)$ be interior to a spindle path of S_{xy} and $z \in V(T)$. Also, let z' be the vertex of T nearest z in the path from x to y in T . Along the cycle formed by the path from x to y in T and the path from y to x via w , there is a path from w to z' , say via x , where any of the labels $j, i_1, i_2, \dots, i_t$ occurs at most once. Continuing our path outside the cycle in T from z' to z , the only label that could repeat is 1, 2, or 4.

If $wx \in E(G)$, then $d(f(w), f(z)) = d(f(x), f(z)) + 1$, because j is a unique label on our path from w to z . Suppose there is another edge on our path before x . If $t \geq 3$, by (v), either y is a leaf of T , so that i_t occurs exactly once on our path, or x is a leaf of T , so i_1 occurs exactly once or $z = x$. Therefore, with $t \geq 3$, if $d_G(w, z) \geq 2$, then $d(f(w), f(z)) \geq 2$. This also holds if $t = 2$: we only need to inspect when w is a midpoint of a P_4 joining one of the four pairs in P .

By (vi), the subgraph $T \leq G$ embeds as an induced subgraph of $\mathcal{Q}_n$. The distance between interior vertices of different spindle paths is at least the minimum distance between two translates of $\mathcal{Q}_n$ in a direction of j , which is 2. Moreover, f embeds non-adjacent vertices in a spindle path and in T to non-adjacent vertices of $\mathcal{Q}_{n+(t+3)|P|}$. Thus, G is an induced cubical subgraph. $\square$

Chapter 4

Monochromatic Sumsets in Countable Colourings of Abelian Groups

4.1 Introduction

It follows directly from Ramsey's theorem that, whenever the naturals are finitely coloured, there is an infinite set X such that all pairwise sums of distinct elements of X have the same colour. However, if we ask for the sumset $X + X = \{x + y : x, y \in X\}$ to be monochromatic, including terms of the form $x + x$, the answer is no. (Indeed, one can find a 3-colouring where z and λz have different colours if $1.9 < \lambda < 2$.) Surprisingly, it is unknown whether or not this can be achieved with a 2-colouring: this is called Owings' problem [35]. For background on this, see [36].

If one extends from the naturals to the rationals, the answer remains no: there is a 12-colouring of $\mathbb{Q}$ with no infinite sumset monochromatic. What about for the reals? Hindman, Leader and Strauss [36] showed that, for every rational vector space of dimension smaller than $\aleph_\omega$, there is a finite colouring without an infinite monochromatic sumset. This establishes the answer for the reals if we assume CH.

In the other direction, Leader and Russell [37] showed that there *are* vector spaces where the answer is yes: if we finitely colour a rational vector space of dimension $\beth_\omega = \sup\{\aleph_0, 2^{\aleph_0}, 2^{2^{\aleph_0}}, \dots\}$, then there is always an infinite monochromatic sumset. Subsequently, Komjáth, Leader, Russell, Shelah, Soukup and Vidnyánszky [38] proved that this positive result also holds for the reals, under a large cardinal assumption, and finally Zhang [39] completed the picture by removing this assumption: he gave an ingenious argument to show that it is consistent that whenever the reals are finitely coloured there is an infinite monochromatic sumset.

Several of these results have been considered and generalised to finite colourings of abelian groups, not just rational vector spaces. See for example Fernández-Bretón [40] and Fernández-Bretón and Lee [41]. For general background and several new results in this direction see Fernández-Bretón, Sarmiento and Vera [42].

Turning to colourings with countably many colours, there are results about monochromatic structures when we countably colour the finite subsets of a (large) set, going back at least as far as Komjáth and Shelah ([43] and [44]) – see also Komjáth [45] and Fernández-Bretón and Rinot [46] for results about abelian groups. However, for sumsets, the situation is very different to the descriptions above. Indeed, one can never guarantee an infinite sumset, however large the abelian group. This was proved in [42]. But what happens for finite sumsets? The authors of [42] gave an elegant argument, based on the Erdős-Rado theorem [47], to show the following: if we countably colour the direct sum of $\beth_\omega$ copies of $\mathbb{Z}_4$, the cyclic group of order 4, then for any n , there is a set X of size n whose sumset $X + X$ is monochromatic.

They asked if these elements of order 4 are necessary, in the following very strong sense: if G is an abelian group with no elements of order 4, does there always exist a countable colouring of G for which there does not even exist a set X of size 2 with $X + X$ monochromatic? In other words, there do not exist distinct x and y such that the elements $2x, 2y, x + y$ all have the same colour. If true, this would make a very attractive counterpart to the result about direct sums of $\mathbb{Z}_4$. They affirmed this in several cases, e.g. when G is a torsion group, having no elements of infinite order.

Our aim is to prove this in general.

Theorem 4.1.1. *Let G be an abelian group not containing any elements of order 4. Then there is a countable colouring of G for which there do not exist distinct x and y with $\{2x, 2y, x + y\}$ monochromatic.*

4.2 Preliminary Observations

Definition 4.2.1. *Let n be a positive integer, G an abelian group, and $x \in G$. We say that x is divisible by n if there exists $y \in G$ such that $x = n \cdot y$. We say that G is a divisible group if for all possible x and n , x is divisible by n .*

It is well-known that if every element of a divisible group has infinite order, then it is a direct sum of subgroups isomorphic to $\mathbb{Q}$, while a divisible torsion group is a direct sum of subgroups isomorphic to $\mathbb{Z}_{p^\infty}$. Here, $\mathbb{Z}_{p^\infty}$ consists of the p^k -th roots of unity for $k = 0, 1, 2, \dots$ [48]

Lemma 4.2.2. *Let p be a prime number and G an abelian group. Then G embeds into an abelian group G_p in which every element is divisible by p .*

Proof. Let x be an element of abelian group G . Suppose for some prime p , that G has no element y such that $p \cdot y = x$. Forming the group $G \oplus \mathbb{Z}$ and quotienting by the subgroup generated by $(x, -p)$ results in a group that G embeds into via $g \mapsto (g, 0)$ and where $p \cdot (0, 1) = (x, 0)$. Adding an element of this group to itself p -fold gives an element of G .

Let G' be the group obtained from G by performing this operation for every element $x \in G$ that is not divisible by p . Then the group $G \cup G' \cup G'' \cup \dots$ (logically, G is not a subset of G' and so on, so we must identify corresponding elements) has all elements divisible by p . $\square$

Corollary 4.2.3. *Let G be an abelian group with no element of order 4. Then G embeds into a group with no element of order 4 that is divisible by odd numbers.*

Proof. Applying Lemma 4.2.2 to every odd prime p , the resulting group H is divisible by every odd prime, hence by odd integers. By construction, for every $y \in H$, there is an odd multiple of y that corresponds to an element of G . If y had order 4, there would be an order 4 element of G . $\square$

The key idea in our proof is that if G is divisible by odd numbers, then its elements of odd order form a divisible subgroup D . This allows us to make use of Baer's theorem.

Theorem 4.2.4 (Baer [49]). *Let G be an abelian group. If $D \leq G$ is a divisible subgroup, then*

$$G \cong D \oplus (G/D).$$

Proof. By Zorn's lemma, there is a maximal subgroup of G disjoint from $D \setminus \{0\}$. We claim that for this group E , $D + E = G$, which implies the result.

If $x \notin E$, let $E(x) = \{v + n \cdot x \mid v \in E, n \in \mathbb{Z}\}$, so that $E \subsetneq E(x) \leq G$. By choice of E , there must be a nonzero $u \in D$ with $v + n \cdot x = u$, where $v \in E$, $n \in \mathbb{Z}$, $n > 0$. Since D is divisible, we can write $u = n \cdot d$ for some $d \in D$, yielding

$$n \cdot (x - d) \in E.$$

Iterating, if $x - d \notin E$, then $v' + n' \cdot (x - d) = u'$ for some nonzero $u' \in D$, where $v' \in E$, $n' \in \mathbb{Z}$, and we may suppose that $0 < n' < n$. Hence, for some $d' \in D$, $n' \cdot (x - d') \in E$. After finitely many iterations, we obtain that $x \in D + E$, as required. $\square$

4.3 Proof of Theorem 4.1.1

Whenever we have an element of a direct sum $\bigoplus_{i \in I} G_i$, we will always assume that the index set I is linearly ordered. This means that each element x of this direct sum has a *support*, meaning $\{i \in I : x_i \neq 0\}$, and also a *profile*, meaning the ordered finite sequence of the non-zero x_i . Note that if each G_i is countable, and there are only countably many different groups that appear as a G_i (whatever the cardinality of I), then there are only countably many profiles.

Proof of Theorem 4.1.1. Given our abelian group G with no elements of order 4, we will build up our colouring in stages, at each time obtaining stronger and stronger information about a supposed monochromatic $\{2a, 2b, a + b\}$, until we obtain a contradiction. Formally, we will be taking one colouring, then taking the product colouring with the next colouring, and so on.

By Corollary 4.2.3, we may assume G is an abelian group without an element of order 4 that is divisible by odd integers. Let D be the subgroup formed by all the odd-order elements of G . Then D is divisible by odd primes, as well as 2, for if $d \in D$ has order $2n - 1$, then $d = 2 \cdot (nd)$. Hence, D is a divisible subgroup of G , and by Theorem 4.2.4, we can write $G = D \oplus X$, where $X \leq G$. Since D is a divisible torsion group, it is a direct sum of copies of $\mathbb{Z}_{p^\infty}$ for (odd) primes p , see [48].

We start countably colouring G by giving (d, x) a colour that is the profile of d . We claim that if $a = (d, x)$ and $b = (e, y)$ have the property that $\{2a, 2b, a + b\}$ is monochromatic, then in fact, $d = e$. To see this, note that $2d, 2e, d + e$ must have the same profile. Now, if the supports of d and e do not end in the same place, then this is impossible, as if the final entry of the profile of d is t and d ends later than e , then the profile of $2d$ ends with $2t \neq 0$, while the profile of $d + e$ ends with t . Hence, the supports of d and e end in the same place, and continuing this argument, we see that d and e have the same support. Since $2d$ and $2e$ have the same profile, it follows that $d = e$.

Next, let T be the torsion subgroup of X . Every element of T has order 2, since other orders would yield either an element of X of order 4 or an element of X of odd order, both of which are impossible. Moreover, the group X/T is torsion-free, and so it embeds into a group Y that is a direct sum of copies of $\mathbb{Q}$. We now additionally colour by giving (d, x) a colour which is the profile of $\pi(x)$, where π denotes the projection from X to X/T , with the latter viewed as a subgroup of Y . By exactly the same argument as above, if $a = (d, x)$ and $b = (e, y)$ have $\{2a, 2b, a + b\}$ monochromatic, then we must have $\pi(x) = \pi(y)$ (the same argument works because, as with D , Y has no elements of even order).

To summarise, we now know that $d = e$ and that x and y belong to the same coset of T in X , in

other words that $x + T = y + T$. It is here that we are rather fortunate. We claim that in each coset of T , there is at most one element that can be halved. Indeed, suppose that $u + T = v + T$ and that there exist g and h with $2g = u$ and $2h = v$. Then $2(g - h)$ is a non-zero element of T , and so has order 2, which implies that $g - h$ has order 4.

So let us add a 2-colouring of G , by the answer to the question, “can this element be halved?”. Now, $2a$ can certainly be halved, but $2x$ and $x + y$ are distinct elements of the same coset of T (distinct because otherwise we would have $d = e$ and $x = y$, whence $a = b$). So, by the above observation, $x + y$ cannot be halved. Thus $a + b$ is not the same colour as $2a$, and we are done. $\square$

It would be very interesting to know what dimension is needed in the result from [42] about direct sums of copies of $\mathbb{Z}_4$. Assuming CH, is $\beth_\omega$ sharp? It would also be nice to find out just how many elements of order 4 need to be present to ensure a sumset $X + X$ with X of size 2 – in other words, to determine precisely which abelian groups have the property that whenever they are countably coloured, there is such a monochromatic sumset.

Chapter 5

Components of Intersecting Polygons

5.1 Introduction

Croft [50] asked the following question.

Question 5.1.1 (Croft [50]). *What is the maximal number of connected components in the intersection of the polygonal regions determined by an n -gon and an m -gon?*



Figure 5.1: intersecting polygons

A *polygon* is a piecewise linear, not self-intersecting, and closed curve in the plane. Every polygon has a diagonal, so by induction, the n -gonal region can be partitioned into $n - 2$ triangles, the m -gonal region into $m - 2$. The intersection of convex regions is convex, every triangle is convex, and a convex region is either path-connected or empty. Thus, the answer to Question 5.1.1 is at most $(m - 2)(n - 2)$, so the maximum exists. Furthermore, if the m -gon is convex, then there are at most $n - 2$ components, and so we may ask how many components there can be at most.

Question 5.1.2. *What is the maximal number of connected components in the intersection of a region bounded by an n -gon and a convex m -gon?*

In Figure 5.1, the yellow regions are in different connected components if the polygons are viewed as open, whereas if the polygons are closed, there is only one connected component. With closed

polygons, a connected component of their intersection could also be a segment or a point. Yet, as we shall see, the maximal number of connected components in Questions 5.1.1 and 5.1.2 does not depend on whether the polygonal regions are open or closed. In the open case, we call each connected component of their intersection an *overlap*.

Our main result is the following.

Theorem 5.1.3. *Let $f(m, n)$ denote the maximal number of overlaps between an n -gonal region and an m -gonal region. Then*

$$\left\lfloor \frac{m}{2} \right\rfloor \cdot \left\lfloor \frac{n}{2} \right\rfloor \leq f(m, n) \leq \left\lfloor \frac{m}{2} \right\rfloor \cdot \frac{n}{2} + \frac{m}{2}$$

Proving the upper bound in Theorem 5.1.3 involves the following lemma.

Lemma 5.1.4. *If no vertex of a polygonal overlap is a vertex of the intersecting m -gon and n -gon, then the overlap has an even number of sides.*

This key idea enables us to determine the maximum precisely when one of the regions is convex.

Theorem 5.1.5. *Suppose we intersect an n -gonal region with a convex m -gonal region. The maximal number of overlaps is then $\min(\left\lfloor \frac{m+n-2}{2} \right\rfloor, n-2)$.*

It is false that $f(m, n) = \left\lfloor \frac{m}{2} \right\rfloor \cdot \left\lfloor \frac{n}{2} \right\rfloor$, as $f(5, 3) = 3$. However, we conjecture this to be nearly true.

Conjecture 5.1.6. $f(m, n) = \left\lfloor \frac{m}{2} \right\rfloor \cdot \left\lfloor \frac{n}{2} \right\rfloor + c_{m,n}$, where $c_{m,n} = 1$ if m and n are odd and 0 if not.

It would be interesting to confirm this conjecture up to a constant, like Ackerman, Keszegh, and Rote [51], who did so for the number of intersection points of an m -gon and n -gon.

5.2 Discussion of connected components versus overlaps

First, we observe that every overlap is an open polygonal region whose boundary is contained by the n -gon or m -gon. This holds because in their triangulations, the intersections of the triangle parts are glued along their diagonal borders. Thus, every vertex of an overlap is an intersection point of the two polygons or a vertex of either.

When intersecting a closed n -gonal and m -gonal region, if the number of connected components is less than for the open versions of the polygons, then the boundaries of two overlaps meet. If it is more, then there is a connected component formed by a segment or a point.

Given a side s , we translate its line by a perpendicular vector, inwards or outwards to its polygon, without moving across a point where another two side lines meet. Such a move does not change the number of overlaps, while decreasing the number of points where three side lines meet.

If the boundaries of two overlaps meet, it is at a vertex of one of the overlaps. Hence, if s is a side that contains that vertex, moving s inwards removes the meeting point. After finitely many such moves, different overlaps form different connected components.

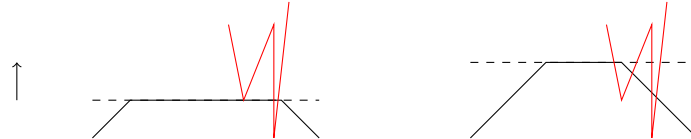


Figure 5.2: before and after a move

If there is a connected component formed by a segment or a point, it is due to a side line of either polygon that coincides or a vertex of one polygon that falls upon the other. By moving the relevant sides outwards, a new overlap is formed at each such connected component, bounded by four sides in the case of the segment, and three or four sides in the case of a point.

Hence, it is clear that if the closed regions of two polygons are considered rather than the open regions, then the maximal number of connected components can be no less and no more.

5.3 Three motivating examples

Our conjectures to the answer to Question 5.1.1 are motivated by special cases where $m, n \leq 5$.

In the case $m = 4$ and $n = 4$, the number of overlaps is no greater than $2 \cdot 2 = 4$, because each quadrilateral is made up of two triangles. Four overlaps can be attained (Figure 5.3), so $f(4, 4) = 4$.

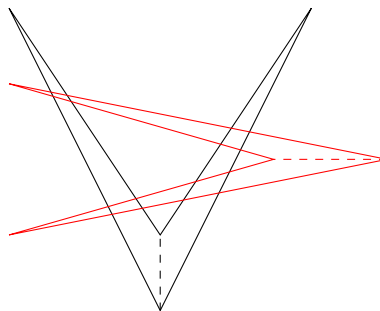


Figure 5.3: $f(4, 4) = 4$

In the case $m = 5$ and $n = 3$, upon triangulating the pentagon, at most $3 \cdot 1$ overlaps emerge. Moreover, as many as three overlaps can be attained. Hence, $f(5, 3) = 3$.

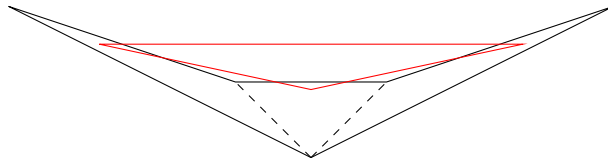


Figure 5.4: $f(5, 3) = 3$

In the case $m = 5$ and $n = 5$, we can simultaneously use the previous two constructions. We add an edge to one of the 4-gons near its concave angle and a vertex to the other near the new edge to form an extra overlap, see Figure 5.5. Thus, we see that $f(5, 5) \geq 5$.

On the other hand, our proof that no more than 5 overlaps are possible is more subtle. If both pentagon regions can be partitioned into two convex regions, then at most $2 \cdot 2 = 4$ overlaps can occur. If not, one of the pentagons has three sides along a concave arc, such as on Figure 5.4, composed of two outer triangles and a middle triangle.

If an outer triangle intersects the other pentagon in three overlaps, then that pentagon is of the same shape. Depending on which vertex of the outer triangle straddles the other pentagon's middle, the interior diagonal side of the outer triangle borders two partial overlaps or there are only three overlaps in total. Either way, the number of overlaps in the outer triangle is at most $1 + \frac{1}{2} + \frac{1}{2} = 2$, if we divide an overlap that meets an interior diagonal into two halves or three thirds.

As for the middle triangle, if it contains at most $1 + \frac{1}{2} + \frac{1}{3}$ overlaps, then there are fewer than six overlaps. Like the outer triangles, it contains no more than two overlaps. It could contain two, either because the other pentagon meets no inner diagonal and at most four overlaps are formed, or because one of the outer triangles meets only one overlap. Thus, by inspection, $f(5, 5) = 5$.

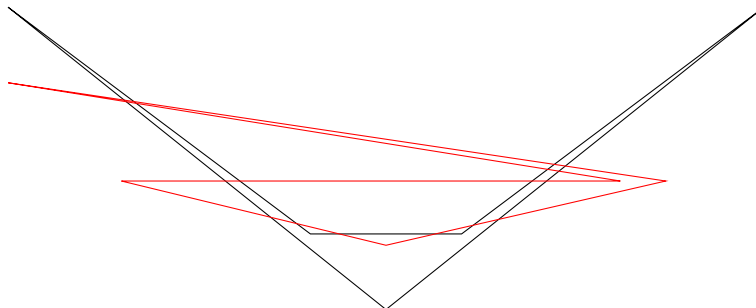


Figure 5.5: construction for $f(5, 5) = 5$

5.4 Proof of Theorem 5.1.3

For the lower bound in Theorem 5.1.3, we intersect $\lfloor \frac{m}{2} \rfloor$ sawteeth with $\lfloor \frac{n}{2} \rfloor$ sawteeth, such as in Figure 5.6, taking care for the first and last side to join up. This yields $\lfloor \frac{m}{2} \rfloor \cdot \lfloor \frac{n}{2} \rfloor$ overlaps, which can be improved by 1 if m and n are odd: we could form a side near a concave angle of one sawtooth and introduce a vertex of the other nearby to see an extra overlap. For instance, the polygons on Figure 5.5 can be produced from the polygons on Figure 5.3 in this way.

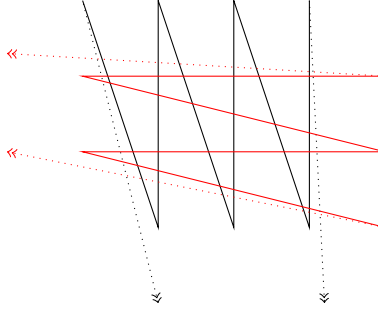


Figure 5.6: construction for Theorem 5.1.3

For the upper bound, we notice that a typical overlap has an even number of sides.

Proof of Lemma 5.1.4. The key is to consider two consecutive sides of an overlap, each of which are along the boundary of the m -gon or the n -gon. If two consecutive sides belong to the m -gon, then the vertex where they meet must be a vertex of the m -gon, and similarly for the n -gon. Otherwise, provided that an overlap does not contain such a vertex, its consecutive sides belong only to the m -gon or only to the n -gon, alternating between the two. Hence, this overlap has an even number of sides along its boundary. $\square$

One side of an n -gon intersects an m -gon in at most m points, which form at most $\lfloor \frac{m}{2} \rfloor$ sides where they overlap. Hence, there are at most $n \cdot \lfloor \frac{m}{2} \rfloor$ sides of an overlap in the n -gon, and similarly for the m -gon. Observing this, we see there are at most $m \cdot n$ sides of overlaps, each with at least four sides, excepting at most $m + n$ with a vertex of either polygon. Thus, Lemma 5.1.4 affords us an upper bound of $\frac{m \cdot n}{4} + m + n$, which we improve as follows.

Suppose there are F overlaps between two regions bounded by the m -gon and the n -gon. Let there be F_k of these with exactly k sides belonging exclusively to the n -gon. If an overlap has two consecutive sides on the m -gon, then they meet at a vertex of the m -gon. We may assume overlaps are disjoint, so each vertex of the m -gon occurs at most once this way. If every side of an overlap

is a side of the m -gon, then all its vertices are an m -gon vertex. If all but one side of an overlap is a side of the m -gon, then it has at least one m -gon vertex. Thus, $3F_0 + F_1 \leq m$.

By double-counting the overlap sides along a side of the n -gon, we see that

$$\sum_{k \geq 0} kF_k \leq n \cdot \left\lfloor \frac{m}{2} \right\rfloor.$$

Adding to this the inequality $3F_0 + F_1 \leq m$, the coefficients of F_0 , F_1 , and F_k for $k \geq 2$ are all equal to 2 or more, so we gather that

$$2F \leq n \cdot \left\lfloor \frac{m}{2} \right\rfloor + m.$$

Dividing by 2 implies the upper bound $f(n, m) \leq \left\lfloor \frac{m}{2} \right\rfloor \cdot \frac{n}{2} + \frac{m}{2}$.

5.5 Proof of Theorem 5.1.5

With regard to Figure 5.4, if a concave arc of $(n - 2)$ sides is formed over a V shape, the resulting n -gon is intersected in m overlaps by a convex m -gon with vertices near the midpoints of the sides along the concave arc, where $m = n - 2$. If m is increased, there still can be $n - 2$ overlaps. If n is increased by 2, then a tooth of two segments can be added (Figure 5.7), so there can be $\left\lfloor \frac{n-(m+2)}{2} \right\rfloor$ more overlaps. We are going to show that this arrangement is optimal for the number of overlaps.

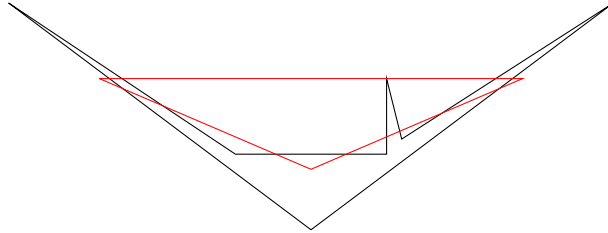


Figure 5.7: appending a tooth

As the n -gon can be partitioned into $n - 2$ triangles, each of which overlap the convex m -gon at most once, there can be no more than $n - 2$ overlaps. Let us further prove that there can be no more than $m + \left\lfloor \frac{n-(m+2)}{2} \right\rfloor = \left\lfloor \frac{m+n-2}{2} \right\rfloor$ overlaps in the case $n \geq m + 2$.

In this problem, a side of the n -gon forms at most one overlap side when intersecting the m -gon, so

$$\sum_{k \geq 0} kF_k \leq n,$$

where F_k is the number of overlaps with k sides along the n -gon. Since $3F_0 + F_1 \leq m$, we deduce

$$2F \leq m + n.$$

Our strategy is to improve this bound from $m + n$ to $m + n - 2$. We single out sides of the n -gon which belong to its convex hull, of which there are at least two. Should these fail to meet the m -gon, clearly $\sum_{k \geq 0} kF_k \leq n - 2$. Similarly, if the m -gon has two vertices outside the n -gonal region, then $3F_0 + F_1 \leq m - 2$. If a side of the m -gon meets the convex hull of the n -gon, then it has a vertex outside the convex hull, and if this vertex is unique, then it can be moved inside the n -gon without changing the number of overlaps, reducing to the former case. Therefore,

$$2F \leq m + n - 2.$$

It follows that no more than $\lfloor \frac{m+n-2}{2} \rfloor$ overlaps can be attained.

Chapter 6

Independence on Toroidal Queens Graphs

6.1 Introduction

The sport of chess is played on an 8×8 board, where various figurines occupy unique fields and move between them. A king, to begin with, moves from its field onto any of the eight fields that share a corner or an edge with it. A bishop and a rook move diagonally and horizontally or vertically, respectively. A queen can move a bishop move or a rook move – parallel to a king move, but unbounded.

A number of journals are devoted to investigating the game theoretic aspects of chess, but we are going to relate chess to graph theory [52]. Consider a figurine graph whose nodes are the fields of a chess board, joined by an edge if a certain figurine, such as a queen, can move between those fields.¹ For such a graph as the queen graph, the inspection of certain graph parameters, e.g. the domination and independence numbers, could interest chess players and combinatorialists alike.

While there are $2n - 1$ diagonals in either direction on the $n \times n$ board, if moving across the boundary is permitted, as though the board were $\mathbb{Z} \times \mathbb{Z}$ modulo n , there are only n diagonals. We identify the fields of this modular, toroidal board with $\mathbb{Z}_n^2$ [53]. It is well-known that n queens can be placed without conflict on the $n \times n$ board [54], or in other words, that there is an independent set of size n in the $n \times n$ queen graph. However, an independent set of n queens exists on the toroidal board $\mathbb{Z}_n \times \mathbb{Z}_n$ if and only if $n \equiv \pm 1 \pmod{6}$ [55].

Although the independence number of the queen graph on $\mathbb{Z}_n^2$ is known [57], we may further

¹We remark that in chess, this relation is not symmetric for pawn moves, but aside from that awkward case, a simple graph is obtained.

inquire into what happens when $d = 2$ is replaced with a higher dimensionality $d > 2$. In order to do that, we must specify how a queen moves on $\mathbb{Z}_n^d$. In this regard, we mention three different possible definitions. Firstly, a queen could move like a two-dimensional queen while any $n - 2$ coordinates are fixed, which we refer to as a plane queen. Secondly, a queen could move along a hyperplane with normal vector in $\{-1, 0, 1\}^d$ [56], to be known as a dual queen. Thirdly, a queen could move parallel to a nonzero vector in $\{-1, 0, 1\}^d$ [53], which we simply call queen. For the first and third versions of queen, we conjecture there is a placement of $n^{d-2}(n - O(1))$ queens without conflict in $d \geq 2$ dimensions, which we prove if all prime divisors of n are at most $2^{\lceil \frac{d+1}{2} \rceil}$. Furthermore, we give the first known impossibility proof concerning a queen in $d \geq 3$ dimensions, establishing that n^2 queens cannot be placed on $\mathbb{Z}_n^3$ without conflict if $n \equiv \pm 5, \pm 10 \pmod{25}$.

6.2 Constructions

We now turn to proving lower bounds for the possible size of an independent subset of the queen graph on $\mathbb{Z}_n^d$, meaning that fields are connected with an edge in this graph if they represent a queen move. The precise maximum for $d = 2$ was found, with proof, by Monsky [57], whose construction we shall present in detail, for sake of completeness. For $d = 3$, Klarner [58] showed that n^2 independent queens on $\mathbb{Z}_n^d$ are attainable if n is coprime to all the primes up to $2^d - 1 = 7$, and like Van Rees [59], we shall extend this to $d > 3$ dimensions.

Ostensibly, due to limitations imposed by the journal and its format, Monsky's examples were unchecked, an omission that was rationalised by declaring this to be "tedious" [57]. Nevertheless, all constructions therein described are slight modifications of an example where the queens occupy the fields $(t, 2t) \in \mathbb{Z}_n^2$ or the fields $(t, 3t) \in \mathbb{Z}_n^2$, where t ranges across all values $t \in \mathbb{Z}_n$. In such a basic example, the fields given by values of t with difference $\frac{n}{2}$ could be in conflict if n is even – this is remedied by suitably translating the second half of the fields. As we are going to see, it is the analogous corrections for conflicts with t values differing by $\frac{n}{3}$ or $\frac{n}{4}$ that may deter with tedium.

Theorem 6.2.1 (Monsky [57]). *There is an independent set of n queens on $\mathbb{Z}_n^2$ if $n \equiv \pm 1 \pmod{6}$, of $n - 1$ queens if $n \equiv \pm 2 \pmod{12}$, and $n - 2$ queens otherwise.*

Proof. Firstly, let $n \equiv \pm 1 \pmod{6}$, meaning that n is coprime with 2 and with 3. Consider the fields labelled with $(t, 2t) \in \mathbb{Z}_n^2$, where $t \in \mathbb{Z}_n$ is a parameter. We claim that if queens are placed on these fields, they will form an independent set.

Notice that when a queen moves from (x, y) by $(0, 1)$, the value of x is invariant. Similarly, moving

by $(1, 0)$, the value of y is invariant. When moving by $(\pm 1, \pm 1)$, the value of $\pm x \mp y$ is invariant. Hence, if two queens are in conflict, it is because the values of x , of y , of $x + y$, or of $-x + y$ agree on the fields they each occupy.

Returning to the fields $(x, y) = (t, 2t)$, the values of x are precisely all possible $t \in \mathbb{Z}_n$. The values of y , of $x + y$, and of $-x + y$ equal $2t$, $3t$, and t , respectively. These are pairwise distinct because n is coprime with 2 and with 3. Hence, the queens cannot be in conflict, as claimed.

Secondly, let $n \equiv \pm 2 \pmod{12}$. Since n is even, $n = 2m$, where $m \equiv \pm 1 \pmod{6}$. Consider

$$(x, y) = (t, 3t), \quad 0 \leq t \leq m - 1 \quad \text{and} \quad (x, y) = (t, 3t + 3), \quad m \leq t \leq 2m - 2.$$

Clearly, the x values are distinct. The values of y are distinct because n is coprime with 3. The values of $-x + y$ are the even $2t$ for $0 \leq t \leq m - 1$ and the odd $2t + 3$ for $m \leq t \leq 2m - 2$, which are distinct, because the integers $0, 2, \dots, n - 2$ are distinct modulo n . The values of $x + y$ are the even $4t$ for $0 \leq t \leq m - 1$ and the odd $4t + 3$ for $m \leq t \leq 2m - 2$, which are also distinct, because the integers $0, 4, \dots, 2n - 4$ cannot differ by n , which is not a multiple of 4. Hence, if queens are placed on these $n - 1$ fields, they will not be in conflict.

Thirdly, let $n \equiv \pm 3 \pmod{12}$, of the form $n = 6m + 3$. The fields

$$(x, y) = \begin{cases} (t, 2t), & 0 \leq t \leq m - 1, \\ (t, 2t + 1), & m \leq t \leq 3m, \\ (t, 2t), & 3m + 2 \leq t \leq 4m + 1, \\ (t, 2t - 1), & 4m + 3 \leq t \leq 6m + 2 \end{cases}$$

are in distinct columns. Their y values are congruent with even integers between $(2m + 1) - (6m + 3) = 2(-2m - 1)$ and $2(4m + 1)$ modulo n , by shifting the instances of $2t + 1$ or $2t - 1$ by $-n$. Thus, differing by less than $2n$, they are distinct. The values of $-x + y$ are distinct, given by the integers from 0 to $n - 2$, excepting m . The values of $x + y$ are $3t$ for $0 \leq t \leq m - 1$ and also for $3m + 2 \leq t \leq 4m + 1$, the latter being $n = 3(2m + 1)$ more than $3t$ for $m + 1 \leq t \leq 2m$, where the integers $3t$ for $0 \leq t \leq 2m$ differ by less than n . Also for this reason, the values of $x + y$ that are 1 modulo 3 or 2 modulo 3 are all distinct in $\mathbb{Z}_n$. Thus, $n - 2$ queens can be placed on these fields independently.

Fourthly, let $n = 12m + 4$, where m is a positive integer. One sees that

$$(x, y) = \begin{cases} (t, 3t), & 0 \leq t \leq 2m - 1, \\ (t, 3t + 2), & 2m \leq t \leq 5m - 1, \\ (t, 3t), & 5m + 1 \leq t \leq 6m + 1, \\ (t, 3t + 1), & 6m + 2 \leq t \leq 9m + 1, \\ (t, 3t + 3), & 9m + 2 \leq t \leq 12m + 2 \end{cases}$$

are in distinct columns, with y values given by the multiples of 3 in the intervals $[0, 6m - 3]$, $[18m + 6, 27m + 3]$ (by adding n), $[15m + 3, 18m + 3]$, $[6m + 3, 15m]$ (by subtracting n), and $[27m + 9, 36m + 9]$, which are disjoint and contained in $[0, 3n - 3]$. Since n is coprime with 3, these integers do not differ by a multiple of n , whence they are distinct in $\mathbb{Z}_n$. The values of $-x + y$ are given by even integers in $[0, 4m - 2] \cup [4m + 2, 10m] \cup [10m + 2, 12m + 2]$ and by odd integers in $[12m + 5, 18m + 3] \cup [18m + 7, 24m + 7]$, which are all distinct. Additionally, the values of $x + y$ differ modulo 4, taking on $4t$ with $0 \leq t \leq 2m - 1$ and $5m + 1 \leq t \leq 6m + 1$, which are distinct as $n = 4(3m + 1)$, as well as a range of values $4t + 1$, $4t + 2$, and $4t + 3$. Therefore, the chosen values of (x, y) form an independent set of $n - 2$ queens.

Fifthly, let $n = 12m + 6$, where m is a positive integer. Define

$$(x, y) = \begin{cases} (t, 3t), & 0 \leq t \leq 2m - 1, \\ (t, 3t + 1), & 2m \leq t \leq 6m + 1, \\ (t, 3t), & t = 6m + 2, \\ (t, 3t - 3), & 6m + 4 \leq t \leq 8m + 3, \\ (t, 3t - 4), & 8m + 5 \leq t \leq 12m + 5. \end{cases}$$

A reader who has followed the proof thus far will enjoy checking that $n - 2$ queens placed on the fields described are not in conflict. For the y values, we note how $n = 3(4m + 2)$. Moreover, inspecting the $\mp x + y$ values bears comparison to the second case, because $n \equiv 2 \pmod{4}$.

Sixthly, let $n = 12m - 4$, where m is a positive integer, using

$$(x, y) = \begin{cases} (t, 3t), & 0 \leq t \leq 2m - 2, \\ (t, 3t - 2), & 2m \leq t \leq 5m - 3, \\ (t, 3t), & 5m - 2 \leq t \leq 6m - 3, \\ (t, 3t - 1), & 6m - 2 \leq t \leq 9m - 5, \\ (t, 3t - 3), & 9m - 3 \leq t \leq 12m - 5. \end{cases}$$

The y values are none other than the multiples of 3 in the intervals $[0, 6m - 6]$, $[18m - 6, 27m - 15]$, $[15m - 6, 18m - 9]$, $[6m - 3, 15m - 12]$, and $[27m - 12, 36m - 18]$, respectively, hence distinct.

Now, the values of $-x + y$ are $2t$ or $2t - 2$ for $0 \leq t < \frac{n}{2}$, which are even, and $2t - 1$ or $2t - 3$ for $\frac{n}{2} \leq t < n$, which are odd. Thus, no two $-x + y$ values are the same. As for the values $x + y$, recall how $n = 4(3m - 1)$. It follows that an independent set of $n - 2$ queens exists in this case.

Seventhly, let $n = 12m$, where $m \geq 2$ is an integer, and place queens upon

$$(x, y) = \begin{cases} (t, 3t), & 0 \leq t \leq 3m - 2, \\ (t, 3t + 3), & 3m - 1 \leq t \leq 4m - 3, \\ (t, 3t + 2), & 4m - 1 \leq t \leq 6m - 2, \\ (t, 3t + 1), & 6m - 1 \leq t \leq 9m - 2, \\ (t, 3t - 2), & 9m \leq t \leq 10m - 1, \\ (t, 3t - 1), & 10m \leq t \leq 12m - 1. \end{cases}$$

By distinguishing the y values modulo 3, the $-x + y$ values modulo 2, and furthermore the $x + y$ values modulo 4, it is evident how these queens are not in conflict.

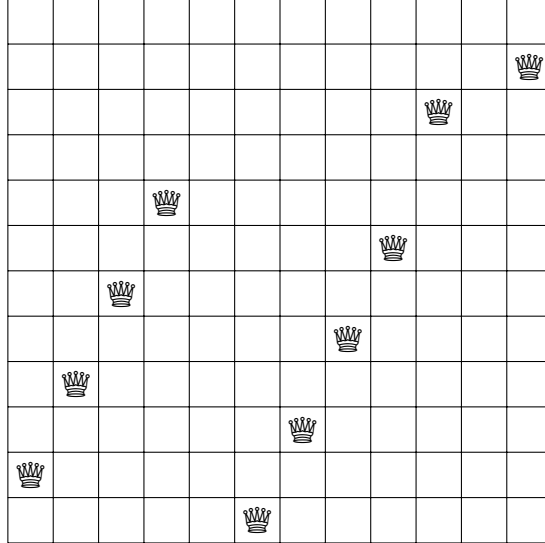


Figure 6.1: 10 queens on $\mathbb{Z}_{12}^2$

Finally, if $n = 4$, place queens on $(0, 0)$ and $(1, 2)$. If $n = 6$, place queens on $(0, 0)$, $(1, 3)$, $(2, 5)$, and $(3, 2)$ – their difference values are 0, 2, 3, 5 and their sum values are 0, 4, 1, 5 in $\mathbb{Z}_6$. If $n = 12$, it is easy to check visually that on Figure 6.1, the set of queens is independent. $\square$

Although our definitions of a plane queen, a dual queen, and a queen coincide in $d = 2$ dimensions, there is a distinction between these in $d > 2$ dimensions, on $\mathbb{Z}_n^d$. Nudelman [56] found an example of n dual queens on $\mathbb{Z}_n^d$ if n is coprime with $(2^d - 1)!$. We give an example of n^{d-1} queens on $\mathbb{Z}_n^d$ if n is coprime with $(2^d - 1)!$ and show that a weaker condition suffices for plane queens.

Theorem 6.2.2. *On the d -dimensional toroidal board $\mathbb{Z}_n^d$, there is an independent set of n^{d-1} queens if n admits no prime factor less than 2^d . Moreover, there is an independent set of n^{d-1} plane queens if n is odd and coprime with $2^j \pm 1$ for $1 \leq j \leq d-1$.*

Proof. Consider placing queens on the following lattice of n^{d-1} fields:

$$\{(t_1, -2t_1 + t_2, -2t_2 + t_3, \dots, -2t_{d-1}) : t_1, \dots, t_{d-1} \in \mathbb{Z}_n\}.$$

Suppose, for the sake of contradiction, that now there are two queens in conflict. By definition, this occurs if and only if their displacement vector is $\mathbf{e} \in \{-1, 0, 1\}^d$ scaled by $\lambda \in \mathbb{Z}_n$; in the case of plane queens, $\mathbf{e}$ has at most two nonzero coordinates. Since the lattice is defined by linear expressions, the displacement between any two fields also has this form. Hence, if two queens on these fields are in conflict, there is a nontrivial solution of

$$(t_1, -2t_1 + t_2, -2t_2 + t_3, \dots, -2t_{d-1}) = \lambda(e_1, e_2, \dots, e_d).$$

In particular, $t_1 = e_1\lambda$, and by solving the recursion $-2t_{l-1} + t_l = \lambda e_l$ ($2 \leq l \leq d$ and $t_d := 0$),

$$t_l = \lambda(e_l + 2e_{l-1} + \dots + 2^{l-1}e_1), \quad l = 1, 2, \dots, d.$$

The final equation reads $0 = \lambda(e_d + 2e_{d-1} + \dots + 2^{d-1}e_1)$, which implies $\lambda = 0$, provided the bracket is coprime with n . Let the first instance of a sign ± 1 in the list $e_d, e_{d-1}, \dots, e_1$ occur at e_{d-i} . The bracket, then, is 2^i times an odd integer bounded by $2^0 + 2^1 + \dots + 2^{d-1} = 2^d - 1$. Indeed, for plane queens, this odd integer is of the form ± 1 or $\pm 1 \pm 2^j$. Hence, if n has no prime factor less than 2^d , then the number in the bracket is coprime with n , and contradiction ensues. For plane queens, it suffices for n to be odd and coprime with $2^j \pm 1$, where $1 \leq j \leq d-1$. $\square$

Remark. While it is possible to place n independent queens on an $n \times n$ board for $n \geq 4$, it is unknown whether n^2 independent queens can be placed on an $n \times n \times n$ board for sufficiently large n . For small cases, Kunt [60] gave a computational proof that if $1 \leq n \leq 14$, this can only be done for $n = 11$ or $n = 13$.

Klarner [58] showed the n^2 queens on $(t_1, t_2, 3t_1 + 5t_2) \in \mathbb{Z}_n^3$ are independent for these values of n , and Van Rees generalised this observation to d dimensions [59], [60]. However, our example in Theorem 6.2.2 adapts well to plane queens, and we will make use of it when constructing independent sets of $n^2 - O(n)$ queens in $\mathbb{Z}_n^3$, where n is not a multiple of 2 or 3.

Conjecture 6.2.3. *For any d , there is an independent set of $n^{d-1} - O(n^{d-2})$ queens on $\mathbb{Z}_n^d$.*

Analogously, Nudelman [56] conjectured that for each dimensionality d , $n = O(1)$ many dual queens can be placed on $\mathbb{Z}_n^d$ without conflict.

By modifying the construction we gave to prove Theorem 6.2.2, we can confirm the claim of Conjecture 6.2.3 in the case that n admits no prime factor less than $2^{\lfloor d/2 \rfloor + 1}$.

Theorem 6.2.4. *For any $d \geq 3$, on the d -dimensional toroidal board $\mathbb{Z}_n^d$, there is an independent set of $n^{d-1} - O(n^{d-2})$ queens if the smallest prime factor of n exceeds $2^{\lfloor d/2 \rfloor + 1}$.*

Proof. For any tuple $\mathbf{t} = (t_1, t_2, \dots, t_{d-1})$ ($t_j \in \mathbb{Z}_n$, $1 \leq j \leq d-1$), let us place a queen upon

$$\mathbf{x} = (t_1, t_2, \dots, t_{d-1}, -2^{d-1}t_1 - 2^{d-2}t_2 - \dots - 2^1t_{d-1} + a(\mathbf{t})),$$

where $a(\mathbf{t})$ is a step function to be specified.

Suppose that some two queens are in conflict along the diagonal of $\mathbf{e} \in \{-1, 0, 1\}^d \setminus \{\mathbf{0}\}$. Clearly, $(e_1, \dots, e_{d-1}) \neq \mathbf{0}$. Now if a were identically zero, then the fields with a queen would satisfy

$$f(\mathbf{x}) := 2^0x_d + 2^1x_{d-1} + \dots + 2^{d-1}x_1 = 0,$$

and so if the queens differ by $\lambda \mathbf{e}$, then $f(\lambda \mathbf{e}) = \lambda f(\mathbf{e}) = 0$ in $\mathbb{Z}_n$. Let m equal

$$f(\mathbf{e}) = 2^0e_d + 2^1e_{d-1} + \dots + 2^{d-1}e_1,$$

a nonzero integer with $|m| \leq 2^0 + 2^1 + \dots + 2^{d-1} = 2^d - 1$.

With $m^* = \gcd(m, n)$, it is the multiples of $\frac{n}{m^*}$ in $\mathbb{Z}_n$ for which $\lambda m \equiv 0 \pmod{n}$. To distinguish between $m^* > 1$ queens in conflict, being displaced by multiples of $\frac{n}{m^*} \mathbf{e}$, we include a perturbation a that varies modulo m^* for each possible value of $m^* > 1$. Since the square of a prime larger than $2^{\frac{d}{2}}$ is more than m , if n has no prime factor up to $2^{\lfloor d/2 \rfloor + 1} = 2^{\lceil \frac{d+1}{2} \rceil}$, then $m^* > 2^{\lfloor d/2 \rfloor + 1}$ is prime.

To determine a modulo $m^* = p$, where $2^{\lceil \frac{d+1}{2} \rceil} < p < 2^d$ and p is a prime divisor of n , we shall divide $[0, n)$ into p equal intervals at multiples of $\frac{n}{p}$. In preparation for this, we consider

$$b(\mathbf{t}) = f(\mathbf{t}, 0) + 2^{\lfloor \frac{d-1}{2} \rfloor} t_1 + 2^{\lfloor \frac{d-1}{2} \rfloor - 1} t_2 + \dots + 2^1 t_{\lfloor \frac{d-1}{2} \rfloor},$$

which has the crucial property that $b(e_1, \dots, e_{d-1})$ is coprime with p if $p \nmid f(\mathbf{e})$, because

$$b(e_1, \dots, e_{d-1}) - f(\mathbf{e}) = -2^0e_d + 2^1e_{\lfloor \frac{d-1}{2} \rfloor} + 2^2e_{\lfloor \frac{d-1}{2} \rfloor - 1} + \dots + 2^{\lfloor \frac{d-1}{2} \rfloor} e_1$$

is bounded by $2^{\lfloor \frac{d-1}{2} \rfloor + 1} \leq 2^{\lceil \frac{d+1}{2} \rceil} < p$ and cannot vanish. Indeed, if it were null, it would follow that $e_1 = e_2 = \dots = e_{\lfloor \frac{d-1}{2} \rfloor} = 0$, so that $p \leq |f(\mathbf{e})| \leq 2^0 + 2^1 + \dots + 2^{d-1} e_{\lfloor \frac{d-1}{2} \rfloor - 1} < 2^{\lceil \frac{d+1}{2} \rceil}$,

a contradiction. Returning to a , we let $a(\mathbf{t}) = a(b(\mathbf{t}))$, such that if $b \bmod n$ is the i -th interval $\left[(i-1)\frac{n}{p}, i\frac{n}{p}\right)$, then $a \equiv i \pmod{p}$. Additionally, for any other prime $q < 2^d$, we let a be divisible by the largest power of q less than 2^d . For each possible value of b , these conditions can be met, by the Chinese Remainder Theorem, with $0 \leq a < (2^d - 1)!$, meaning $a = O(1)$.

Suppose, now, that a conflict occurs because we placed queens at $\mathbf{x}$ and $\mathbf{x} + \lambda \mathbf{e}$. Observe that $f(\mathbf{x}) = a(\mathbf{t})$ and $f(\mathbf{x} + \lambda \mathbf{e}) = a(\mathbf{t}')$ for some tuples $\mathbf{t}$ and $\mathbf{t}'$, whence

$$\lambda f(\mathbf{e}) = a(\mathbf{t}') - a(\mathbf{t}),$$

and we let $f(\mathbf{e}) = m$. The integer $a(\mathbf{t}') - a(\mathbf{t})$ is thus divisible by $m^* = \gcd(m, n)$. Also, both $a(\mathbf{t})$ and $a(\mathbf{t}')$ are divisible by $\frac{m}{m^*}$, which is less than 2^d , for if it were divisible by a prime factor of n , then m would be divisible by two prime numbers that exceed $2^{d/2}$, contradicting $m < 2^d$. It follows that $\lambda_0 = \frac{a(\mathbf{t}') - a(\mathbf{t})}{m}$ is an $O(1)$ integer, and $\lambda \in \{\lambda_0 + j \cdot \frac{n}{m^*} \in \mathbb{Z}_n \mid j = 0, 1, \dots, m^* - 1\}$.

In the case where $\lambda \neq \lambda_0$, $m^* > 1$, so m^* is a prime divisor p of n . With $a(\mathbf{t}') \equiv a(\mathbf{t}) \pmod{p}$, there is an $i \in \{1, 2, \dots, p\}$ such that $b(\mathbf{t}'), b(\mathbf{t}) \in \left[(i-1)\frac{n}{p}, i\frac{n}{p}\right)$. On the other hand,

$$b(\mathbf{t}') - b(\mathbf{t}) = \lambda b(e_1, \dots, e_{d-1}),$$

where λ is multiplied by a number coprime to p . Hence, if $\lambda = \lambda_0 + j \cdot \frac{n}{p}$ and j does not vanish modulo p , then up to an $O(1)$ term, $b(\mathbf{t}') - b(\mathbf{t})$ is a nontrivial multiple of $\frac{n}{p}$. This can be avoided if we remove the queens for which $b(\mathbf{t})$ is within $O(1)$ distance of any $i \cdot \frac{n}{p}$.

In the case where $\lambda = \lambda_0$, $b(\mathbf{t}') - b(\mathbf{t}) = O(1)$, yet $a(b(\mathbf{t}')) \neq a(b(\mathbf{t}))$. By removing the queens for which $b(\mathbf{t})$ is within $O(1)$ distance of any of the jumps of $a(b)$, these conflicts disappear.

Since n is odd, for each value of $t_1, \dots, t_{d-2}$, $b(\mathbf{t}) \in \mathbb{Z}_n$ is surjective in t_{d-1} . Thus, by erasing $O(1)$ of its values, $O(n^{d-2})$ queens are lost, leaving $n^{d-1} - O(n^{d-2})$ independent queens. $\square$

6.3 An impossibility proof

Given an independent set of queens on $\mathbb{Z}_n^d$, restricting to fields with a fixed final coordinate yields an independent set of queens on $\mathbb{Z}_n^{d-1}$ for $d \geq 2$. Hence, at most n times as many queens can be placed on $\mathbb{Z}_n^d$ independently as on $\mathbb{Z}_n^{d-1}$. By induction, the size of an independent set of queens on $\mathbb{Z}_n^d$ is at most n^{d-1} .

By Pólya's result in two dimensions [55], a set of n^{d-1} queens on $\mathbb{Z}_n^d$ cannot be independent if n is a multiple of 2 or 3, where $d \geq 2$. Similarly to Nudelman [56], we exhibit a result in $d \geq 3$ dimensions that shows this also holds if n is divisible by 5 and not by 5^2 .

Theorem 6.3.1. *Let n be a positive integer that is not a multiple of 2 or 3, divisible by 5, and not a multiple of 25. For such n , there is no independent set of n^2 queens on $\mathbb{Z}_n^3$.*

Proof. Suppose that queens are placed on

$$(x_1, y_1, z_1), \dots, (x_{n^2}, y_{n^2}, z_{n^2}) \in \mathbb{Z}_n^3.$$

We claim that for these fields (x, y, z) , the pairs (y, z) attain pairwise distinct values, and similarly for $(x \pm y, z)$ and $(x \pm y, x \pm z)$ with any given signs. This is the case because if two fields gave rise to the same value of a pair, then they would be apart by a multiple of $(1, 0, 0)$, $(1, \mp 1, 0)$, or $(1, \mp 1, \mp 1)$, respectively. Hence, each of these n^2 pairs take on every value in $\mathbb{Z}_n^2$.

Let k and l be positive integer exponents, and consider the sum

$$\sum_{i=1}^{n^2} x_i^k y_i^l.$$

Since (x_i, y_i) takes on every possible value precisely once, this equals

$$\sum_{(x,y) \in \mathbb{Z}_n^2} x^k y^l = \left(\sum_{x \in \mathbb{Z}_n} x^k \right) \left(\sum_{y \in \mathbb{Z}_n} y^l \right).$$

The value of a power sum $S_k = \sum_{x=0}^{n-1} x^k$ can be computed recursively, due to the well-known combinatorial identity $\sum_{x=k}^{n-1} \binom{x}{k} = \binom{n}{k+1}$, which we write as

$$\sum_{x=0}^{n-1} \frac{x(x-1) \dots (x-k+1)}{k!} = \frac{n(n-1) \dots (n-k)}{(k+1)!}.$$

Upon multiplying both sides $k!$, we can expand $x(x-1) \dots (x-k+1)$, the sum of x^k and lower order terms, which, when summed over x , yields the sum of S_k and an integer combination of $S_{k-1}, \dots, S_0$. Since n is not a multiple of 2 or 3, it divides the integer $\frac{1}{k+1} n(n-1) \dots (n-k)$ for $k \leq 3$. Since n is a multiple of 5, for $k = 4$, this expression is $\frac{n}{5}(-1)(-2)(-3)(-4) \equiv -\frac{n}{5} \pmod{n}$. Thus, we deduce that $S_k \equiv 0 \pmod{n}$ if $k \leq 3$ and $S_4 \equiv -\frac{n}{5} \pmod{n}$.

Projecting to $\mathbb{Z}_n$, these considerations motivate us to work with

$$\sum_{i=1}^{n^2} x_i^4 y_i^4 = S_4 \cdot S_4,$$

which does not vanish, provided that n is not divisible by 25. If we use the shorthand $\sum f(x, y, z)$ to mean $\sum_{i=1}^{n^2} f(x_i, y_i, z_i)$, then since pairs of the type $(x \pm y, z)$ and $(x \pm y, x \pm z)$ also attain every possible value,

$$\begin{aligned}\sum y^4(x \pm z)^4 &= S_4^2, \\ \sum (x \pm y)^4(x \pm z)^4 &= S_4^2.\end{aligned}$$

To cancel terms, we add, subtract, and factor:

$$\sum [(x+y)^4 + (x-y)^4] \cdot [(x+z)^4 + (x-z)^4] = 4 \cdot S_4^2, \quad (6.1)$$

$$\sum [(x+y)^4 + (x-y)^4 - 2y^4] \cdot [(x+z)^4 + (x-z)^4 - 2z^4] = 0. \quad (6.2)$$

Since $(x+y)^4 + (x-y)^4 = 2(x^4 + 6x^2y^2 + y^4)$, (6.2) shows $\sum (x^4 + 6x^2y^2)(x^4 + 6x^2z^2) = \sum x^8 + \sum x^6z^2 + \sum x^6y^2 + \sum x^4y^2z^2$ to vanish. Of course, we know that $\sum x^6y^2 = S_6S_2$ vanishes, as well as $\sum x^6z^2 = S_6S_2$ and $\sum x^8 = S_8S_0$. Thus, we learn that $\sum x^4y^2z^2$ vanishes modulo n . Returning to (6.1),

$$\sum (x^4 + 6x^2y^2 + y^4)(x^4 + 6x^2z^2 + z^4) = S_4^2$$

implies that

$$\begin{aligned}& \sum (x^8 + 6x^6z^2 + 6x^6y^2) \\ & + \sum (x^4y^4 + x^4z^4 + y^4z^4) \\ & + \sum (6x^2y^2z^4 + 6x^2y^4z^2 + 36x^4y^2z^2) \equiv S_4^2 \pmod{n}.\end{aligned}$$

Of these three terms, we know that the first term vanishes, and the third vanishes by virtue of $\sum x^4y^2z^2 = 0$ and its variants. The remaining second term is $3 \cdot S_4^2$, whence we arrive at a contradiction, $3S_4^2 = S_4^2$, i.e. $S_4^2 = 0$ in $\mathbb{Z}_n$. $\square$

Our proof illustrates the limitations of an impossibility proof involving sums over all possible values modulo n . If n were a multiple of 7, say, in d dimensions, there would be $\frac{3^d-1}{2}$ equations involving the sum of sixth powers, but far more homogeneous monomials of degree $7(d-1)$ to cancel. For this reason, it would be a breakthrough to determine, even for $d=3$, when there is an independent set of n^{d-1} queens on $\mathbb{Z}_n^d$.

Interestingly, Monsky further proved that there is no independent set of $n-1$ queens on $\mathbb{Z}_n^2$ if n is a multiple of 3 or 4 [57], in which case at most $n(n-2) = n^2 - 2n$ independent queens can be placed on $\mathbb{Z}_n^3$. Compared to this, it would seem unlikely that the upper bound $n^2 - 1$ from Theorem 6.3.1 can be attained.

Chapter 7

Greedy Gossiping

7.1 Introduction

In 1971, Boyd asked the following question [62]. There are n people who each know an item of gossip. In a telephone call, two people share all the gossip they know. How many calls are needed for all of them to be informed of all the gossip?

Boyd's problem, commonly referred to as the Gossiping Problem, was resolved within a year in various ways by Tijdeman [63], Baker and Shostak [64], as well as Hajnal, Milner, and Szemerédi [62]. Bollobás posed it as a challenging starred question for undergraduate students [65]. Thus, it is well-known that $2n - 4$ calls are sufficient for $n \geq 4$, as well as necessary.

In this chapter, we tackle the related question of what happens when the number of calls is restricted to m calls. For the sake of our word problem, we say the dons live in different countries and international telephone calls are very costly. We name this Greedy Gossiping, for the dons shall aim to know as much gossip as possible in total. That is, let each don count how many items of gossip she knows, and let N be the sum total. The maximal value of N , given m , is the following.

Theorem 7.1.1. *Let $n \geq 4$ be an integer and $0 \leq m \leq 2n - 5$. Suppose that n dons each know a unique item of gossip, and in each of m telephone calls, two of the dons communicate all the items of gossip they know and remember it. After the calls, each don counts how many items of gossip she knows. The sum of these numbers, N , is the score of the call sequence.*

The largest possible score of gossiping is achieved by the first m of the following $2n - 3$ calls: dons $2, 3, \dots, n$ call don 1, and then don 1 calls dons $2, 3, 4, \dots, n - 1$.

The original Gossiping Problem prompted a wide array of related questions [67]. Among these, perhaps the question most similar to Greedy Gossiping is Partial Gossiping, where after m calls, it is the fewest items of gossip known by a don that is to be maximised [68]. Further, we also mention three variants of Gossiping. In Broadcasting, information from one don is to reach all the dons, under further constraints [67]. In Perpetual Gossiping, a new piece of information occurs to each don after a given number of calls are made [69]. In Shuffling, each don has a unique object, and a lazy call is an option to exchange the object [70].

In our proof of Theorem 7.1.1, the more the calls, the more difficult the proof is. To begin with, we give a formula for the score we claim to be maximal. In the range $0 \leq m \leq n - 1$, it is easy to see that a greedy strategy, where each subsequent call maximally increases the score, is globally optimal. Thereafter, having solved the case $m = n - 1$, in the range $n - 1 \leq m \leq 2n - 6$, we induct on n , using lemmas due to Baker and Shostak [64]. Finally, if $m = 2n - 5$, we require a lemma by Kleitman and Shearer [66] that, unlike the lemmas of Baker and Shostak, also applies to the version of Gossiping where k dons participate in every call. This raises the question of whether Theorem 7.1.1 can be generalised by letting every call involve k dons, where $k \geq 2$.

The Gossiping Problem has recently been studied in connection with computer science [71], [72]. Accordingly, we refer to a call sequence among n dons as a temporal graph. We mention a few questions that occurred in this recent algorithmic field of research. In a subsequence of a certain call sequence, at least how many calls are needed so that every don hears every gossip? What is the maximum of this number if the call sequence is the edges of a complete graph in some order [71]? Also, inspired by the Travelling Salesman Problem, we could ask, how many in-person meetings are necessary to pass an object along so that every don observes it [72]?

7.2 Solution to Greedy Gossiping

For the sake of proving Theorem 7.1.1, we define the concept of a temporal path. Let the dons $v_1, v_2, \dots, v_n$ make the calls $e_1, e_2, \dots, e_m$, where call e_t takes place at time t . These calls form a temporal graph, because the calls list pairs of dons with specific timestamps. A subsequence of $e_1, e_2, \dots, e_m$ forms a temporal subgraph, and we say it is a temporal walk from u to w if it is of the form $\{u_1, u_2\}, \{u_2, u_3\}, \dots, \{u_{h-1}, u_h\}$ for some dons $u = u_1, u_2, u_3, \dots, u_{h-1}, u_h = w$. A temporal walk is a temporal path if the dons are distinct. Clearly, the item of gossip from don u reaches don w if and only if a temporal walk (omitting loops, a temporal path) from u to w exists.

The optimal value of N

The optimal call sequence described in Theorem 7.1.1 is given by

$$c_t = \begin{cases} \{v_1, v_{t+1}\} & (1 \leq t \leq n-1) \\ \{v_1, v_{t-(n-1)+1}\} & (n \leq t \leq 2n-3) \end{cases}$$

and then $e_t := c_t$ for $t = 1, \dots, m$.

Regarding $(c_t)_{t=1}^{2n-3}$, we claim that after m calls,

- if $0 \leq m \leq n-1$, v_1 knows the gossip from $v_1, \dots, v_{m+1}$, while v_i knows the gossip from $v_1, \dots, v_i$ for $2 \leq i \leq m+1$;
- if $n \leq m \leq 2n-3$, $v_1, \dots, v_{m-n+2}$ know all the gossip, and v_i knows the gossip from $v_1, \dots, v_i$ for $m-n+3 \leq i \leq n$.

By induction, this claim follows from how

- if $1 \leq t \leq n-1$, c_t lets v_1 learn the gossip from v_{t+1} and v_{t+1} learn the items of gossip due to $v_1, \dots, v_t$;
- if $n \leq t \leq 2n-3$, c_t lets v_{t-n+2} learn all items of gossip she did not know previously.

Consequently, the score of $(c_t)_{t=1}^m$ can be characterised as follows:

$$N = \begin{cases} n + (2 + 3 + \dots + (m+1)) & \text{if } 1 \leq m \leq n-1 \\ n^2 - (1 + 2 + \dots + (l-1)) & \text{if } m = 2n-2-l, 1 \leq l \leq n-1 \end{cases}$$

where the latter case describes how for $1 \leq i \leq n-1$, v_{n-i} does not know i pieces of information after the first $n-1$ calls, and the next $n-1-l$ calls inform $v_1, \dots, v_{n-l}$ with every gossip item.

The case $0 \leq m \leq n-1$

In describing the optimal call sequence $(c_t)_{t=1}^m$ and the score N , the case of $0 \leq m \leq n-1$ was relatively simple. To prove its optimality, we simply induct on m .

Lemma 7.2.1. *In call t ($1 \leq t \leq n-1$), denoted e_t , no more than $t+1$ items of gossip are heard.*

Proof. Let us consider the simple graph whose vertices are v_i ($1 \leq i \leq n$) and whose edge set is $\{e_1, e_2, \dots, e_t\}$. If an item of gossip is heard in call t , then it reached one of the dons in e_t via a temporal path. Hence, the sources of the gossip communicated in e_t are limited to the connected

components of the two dons involved. Since a connected graph with at most t edges contains at most $t + 1$ vertices, no more than $t + 1$ items of gossip can be heard in the call e_t . $\square$

From Lemma 7.2.1, we can see that call t increases the score by at most $t + 1$, so if $1 \leq m \leq n - 1$,

$$N \leq n + \sum_{t=1}^m (t + 1).$$

For equality to hold in this bound, the graph of calls until call t must be connected for $1 \leq t \leq m$. Since the graph of calls until call $t - 1$ is connected, e_t is a call between a new don and a don who must know every item of gossip from the dons in previous calls, a don who is one of the dons in call $t - 1$. While such a call can be arranged in multiple ways, it is precisely one that increases the score maximally in call t ($t = 1, 2, \dots, m$). Thus, if we arrange the first $m \leq n - 1$ calls greedily, then up to renaming the dons, the sets of gossip items known to each don are uniquely determined.

The case $n - 1 \leq m \leq 2n - 6$

Having settled the case $m = n - 1$, we show the maximal score is $N = n^2 - (1 + 2 + \dots + (l - 1))$ when there are $m = 2n - 2 - l$ calls and $4 \leq l \leq n - 1$. If $n = 4$, there is nothing to prove.

Suppose Theorem 7.1.1 holds for n dons with $n - 1 \leq m \leq 2n - 6$ calls. Let $n + 1$ dons make $m = 2n - l$ calls, where $4 \leq l \leq n$. If $l = n$, it is the case that $m = n$, so we are done. If $l < n$, the induction step is made possible by the following observation [64].

Lemma 7.2.2 (Baker and Shostak [64]). *Suppose that v_{n+1} hears her own item of gossip, i.e. there are $n + 1$ dons whose call sequence includes a (nontrivial) temporal walk from v_{n+1} to v_{n+1} . Under this premise, it is possible to arrange two fewer calls among $v_1, \dots, v_n$ without communicating any less information among them.*

Proof. As a preliminary, we remark that one less call can always be arranged. Suppose the calls of v_{n+1} are with dons $u_1, u_2, \dots, u_d$, in this order. Instead of these calls, for $1 \leq i \leq d - 1$, let u_i call u_{i+1} at the time she would call v_{n+1} . This way, if a temporal path with endpoints among $v_1, \dots, v_n$ includes $\{u_i, v_{n+1}\}$ and $\{v_{n+1}, u_j\}$, $i < j$, then these two edges can be replaced with a temporal walk $u_i u_{i+1} \dots u_{j-1} u_j$ to form a temporal walk not involving v_{n+1} .

Now suppose there is a temporal walk from v_{n+1} to v_{n+1} . Hence, between some two calls $\{v_{n+1}, u_p\}$ and $\{v_{n+1}, u_q\}$, $p < q$, there is a (possibly trivial) temporal path P from u_p to u_q . Informally, on our preliminary temporal walk $u_1 u_2 \dots u_d$, rather than using $u_p u_{p+1} \dots u_q$, we walk from u_p to u_q

along P . However, for $p < i < q$, at the time u_i would call v_{n+1} , she calls the don at the endpoint of the path P up to that time. Given a temporal path with endpoints among $v_1, \dots, v_n$ including $\{u_i, v_{n+1}\}$ and $\{v_{n+1}, u_j\}$, $i < j$, these two edges can be replaced with a subpath of our temporal walk from u_1 to u_d , possibly with a replacement call from u_i or u_j to begin or end with. In this way, we reduce the number of calls by 2 without changing the information communicated between two of the dons $v_1, \dots, v_n$. $\square$

The upshot of Lemma 7.2.2 is that if there is a call sequence of $2n - l$ calls ($4 \leq l \leq n - 1$) among $n + 1$ dons that scores N , then there is a call sequence of $2n - 2 - l$ calls among n dons that scores at least $N - (2n + 1)$. Indeed, the don v_{n+1} spreads her item of gossip to at most n other dons and knows at most $n + 1$ items of gossip in the end. Therefore, by induction,

$$N - (2n + 1) \leq n^2 - (1 + 2 + \dots + (l - 1)),$$

$$N \leq (n + 1)^2 - (1 + 2 + \dots + (l - 1)).$$

As we are done in this case, we may suppose that no don hears her own item of gossip. The next observation, also from [64], shows the ramifications of this premise.

Lemma 7.2.3 (Baker and Shostak [64]). *Suppose that v_i does not hear her own gossip. Let a_i be the number of dons who know the gossip of v_i and let v_i know b_i items of gossip. Further, let r_i be the number of calls that neither inform v_i nor pass on v_i 's gossip, and let s_i be the number of calls with v_i . Then*

$$a_i + b_i \leq m + 2 + s_i - r_i.$$

Proof. Highlight all the calls where v_i 's gossip was known by only one of the dons. There are at least $a_i - 1$ such calls, of which at least $a_i - 1 - d_i$ are not a call with v_i , so these calls are in a temporal path from v_i of more than one call. Since there is no temporal walk from v_i back to v_i , it follows that these calls cannot be in a temporal path to v_i . Such temporal paths include at least $b_i - 1$ edges, because they connect b_i vertices, including v_i . Altogether, we have

$$(a_i - 1 - s_i) + (b_i - 1) + r_i \leq m,$$

which rearranges to the claimed inequality. $\square$

The score, $N = \sum_{i=1}^n b_i$, counts the number of pairs (i, j) for which v_i hears an item of gossip from v_j , and so $N = \sum_{j=1}^n a_j$. Similarly, by double-counting the number of participants in the m

calls, we obtain $2m = \sum_{i=1}^n s_i$. Therefore, we can sum both sides of Lemma 7.2.3 to find

$$N + N \leq n(m + 2) + 2m - \sum_{i=1}^n r_i.$$

We are done, because $\sum_{i=1}^n r_i \geq 0$ and $(n + 2)m + 2n \leq 2n^2 - 2(1 + 2 + \dots + (l - 1))$. Indeed, the latter inequality is equivalent to $(n + 2)(2n - 2 - l) + 2n \leq 2n^2 - l(l - 1)$, which in turn rearranges to $(l - 4)(n - 1 - l) \geq 0$.

The case $m = 2n - 5$

Let us now tackle the case $m = 2n - 5$. Since the previous argument, up to Lemma 7.2.3, is valid when $l = 3$, without loss of generality, no don hears her own item of gossip. Our next observation is that a swap of disjoint consecutive calls does not change who knows which gossip in the end.

Lemma 7.2.4 (Kleitman and Shearer [66]). *If the dons make $m \geq n - 1$ calls, then there is a rearrangement of the calls that ultimately communicates the same information where either some don is not involved in the first $n - 2$ calls or e_{n-2+i} is never disjoint from each of $e_{n-2}, e_{n-1}, \dots, e_{n-3+i}$ for $1 \leq i \leq m - (n - 2)$.*

Proof. The key idea, due to Hajnal, Milner, and Szemerédi [62], is that if two consecutive calls are disjoint, then by swapping them, the items of gossip known by the participants after the calls does not change. In particular, suppose that e_{j+i} is disjoint from $e_j, e_{j+1}, \dots, e_{j+i-1}$. Then, by induction on i , there are i such swaps that rearrange $e_j, e_{j+1}, \dots, e_{j+i}$ to $e_{j+i}, e_j, \dots, e_{j+i-1}$.

Let us keep track of the connected components in the graph G with vertices $v_1, \dots, v_n$ and edges given by the first $n - 2$ calls. At least two of these components are trees, because a connected graph that is not a tree has at least as many edges as vertices. If none of the trees is an isolated vertex, then by making swaps between calls in disjoint components, we can let e_{n-2} be the final call in the smallest of the trees, T_1 , and e_{n-3} be the final call in the second smallest of the trees, T_2 .

Now if there is a positive integer i for which e_{n-2+i} is disjoint from $e_{n-2}, \dots, e_{n-3+i}$, then the calls can be rearranged so that the first $n - 1$ calls are $\dots, e_{n-3}, e_{n-2+i}, e_{n-2}$. Thus, G changes, as T_1 is disconnected into subtrees T'_1 and T''_1 , and then some two connected components are joined with an edge. This decreases $|T_1| + |T_2|$ if two of T'_1, T''_1 , and T_2 are among its components. Otherwise, e_{n-2+i} joins two of T'_1, T''_1 , and T_2 , so $|T_1| + |T_2|$ stays the same or decreases.

If e_{n-2+i} is from T'_1 to T''_1 , then with two swaps, we can rearrange the calls $e_{n-3}, e_{n-2+i}, e_{n-2}$ to $e_{n-2+i}, e_{n-2}, e_{n-3}$. This way, G will include two components that are subtrees of T_2 , and so

$|T_1| + |T_2|$ decreases by at least $|T_1|$. If e_{n-2+i} is from T_2 to T'_1 , say, then T''_1 will be a smaller tree component in G . Therefore, $2^{|T_1|+|T_2|} + 2^{|T_1|}$ is a monovariant, and at its minimum over all possible rearrangements of the calls, either an isolated vertex occurs or no call e_{n-2+i} can be swapped with $e_{n-2}, e_{n-1}, \dots, e_{n-3+i}$. $\square$

In the case where some don v_i has not called until e_{n-2} , at most $n - 2$ dons learn her gossip, and so $a_i \leq n - 2$. In the case where the last $n - 2$ calls are connected as a graph, there is an isolated vertex representing a don v_i that only learns from the first $n - 3$ calls, whence $b_i \leq n - 2$.

Notice that by reversing the calls $e_1, \dots, e_m$ to $e_m, \dots, e_1$, all temporal paths are also reversed, and so a_i and b_i are interchanged. Hence, without loss of generality, $a_i \leq n - 2$ for some i .

Since $a_i \leq n$ for all i , we obtain $N = \sum_{i=1}^n a_i \leq n^2 - 2$. Our objective is to prove that $N \leq n^2 - 3$, which we may do by ruling out the case of equality, i.e. $a_i = n$ for all i but one with $a_i = n - 2$. For the sake of contradiction, let us assume that is the case.

Lemma 7.2.5. *For each don, let us consider who they called first. Suppose the call $\{v_1, v_2\}$ came first for v_1 , but $\{v_2, v_3\}$ came first for v_2 . Then $a_1 < n$, unless v_3 hears her own item of gossip.*

Proof. If $a_1 = n$, there must be a temporal path from v_1 to v_3 . The first call in this path is $\{v_1, v_2\}$ or a later call, and so it occurs after $\{v_2, v_3\}$. Therefore, by appending $\{v_2, v_3\}$ and removing or inserting $\{v_1, v_2\}$, we obtain a temporal walk from v_3 to v_3 . $\square$

If a call is the first for both callers, we say it is a mutual first call, and similarly for final calls.

Notice that if $\{v_1, v_2\}$ is a mutual first call, then afterwards, whoever knows v_1 's gossip also knows v_2 's gossip. Hence, for a unique i with $a_i = n - 2$, the first call is not mutual.

By Lemma 7.2.5, whenever $a_i = n$, the first call of don v_i has to be mutual. Thus, there are $\frac{n-1}{2}$ mutual first calls. As for mutual final calls, we would be done if there were three distinct vertices whose final call is not mutual, with $b_i \leq n - 1$ thrice. Hence, for parity reasons, $\frac{n-1}{2}$ final calls are mutual. The mutual first and mutual final calls are different, or else there would be two dons disconnected from the others, a contradiction. Thus, there are $\frac{n-1}{2}$ mutual first calls, $\frac{n-1}{2}$ mutual final calls, and $n - 4$ miscellaneous calls.

Lemma 7.2.6. *If these miscellaneous calls do not isolate a vertex, then $r_i \geq 1$ for all i .*

Proof. The $n - 4$ miscellaneous calls determine a graph on $v_1, \dots, v_n$ with at least four connected

components. Any temporal path beginning at v_i or ending at v_i could contain miscellaneous edges from the component of v_i , her first caller, or her final caller. Therefore, a miscellaneous edge in a fourth component would not be involved in communicating to v_i or from v_i . $\square$

By Lemma 7.2.3, $a_i + b_i \leq m + 2 + s_i - r_i$, which we rewrite in the form

$$s_i + (n - a_i) + (n - b_i) \geq 3 + r_i.$$

If it were the case that $r_i \geq 1$ for all i , then summing would yield $2m + 2(n^2 - N) \geq 4n$, $N \leq n^2 - 5$. Otherwise, there is a don v_j whose vertex is isolated in the graph of miscellaneous calls. The only calls involving v_j are her first and final calls, and so $s_j = 2$.

Collecting the facts, we find a unique don whose first call is not mutual, say v_1 , so that $a_1 = n - 2$, as well as a unique don, say $v_{j'}$, whose final call is not mutual, which implies $b_{j'} < n$. Since v_j cannot be v_1 , it follows that $a_j = n$. Using Lemma 7.2.3, we find $2 + 0 + (n - b_j) \geq 3$, so $b_j < n$. However, since the same information is known by mutual final callers, the dons in the final call of v_j and also $v_{j'}$ are now three instances of $b_i \leq n - 1$. Thus, we deduce that $N \leq n^2 - 3$.

This completes the proof of Theorem 7.1.1.

7.3 Conclusion

Although the bulk of our proof built on different solutions to the Gossiping Problem, the key idea for the subcase $m = 2n - 5$ is the heart of a more general proof that works if calls are between k people, where $k \geq 2$ is fixed. Hence, a logical next question for Greedy Gossiping is whether the same phenomenon occurs if the calls form a temporal k -uniform hypergraph.

Question 7.3.1. *Let n dons gossip in m conference calls of k people. Suppose that m is less than the minimal number of calls needed for all the dons to know all the gossip. Is the score, i.e. the sum of how many gossip items are known to each don, maximised by a greedy call strategy?*

If $1 + m(k - 1) \leq n$, then for connectivity reasons, call t communicates no more than $1 + t(k - 1)$ items of gossip ($1 \leq t \leq m$). Thus, call t increases the score by at most $(k - 1)$ times as much, and this can be realised if each new call involves $k - 1$ new dons and a don from the previous call, so the answer to Question 7.3.1 is yes in this case. However, in the remaining case of $\frac{n-1}{k-1} < m$ (where $m < \lceil \frac{n-k}{k-1} \rceil + \max(\lceil \frac{n}{k} \rceil, \lceil \frac{n-k}{k-1} \rceil)$, see [66]), our question is open.

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